How Data-Driven Decision-Making Influences Marketing Effectiveness in the FMCG Industry: A Critical Qualitative Study

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Abstract

Fast-Moving Consumer Goods (FMCG) companies have invested heavily in data-driven decision-making (DDDM), yet a persistent gap remains between data availability and meaningful marketing outcomes. This paper examines how DDDM shapes marketing effectiveness in FMCG, asking whether analytics investment translates into superior performance across revenue growth, customer base development, portfolio expansion, and marketing team capability.

The study adopts the interpretivist approach and a qualitative secondary data analysis of the literature drawn from peer-reviewed journal articles, industry reports, published FMCG case studies and company performance reports from FMCG companies. Reflexive thematic analysis was done manually following Braun and Clarke's (2006; 2019) six-phase framework, and four themes were developed.

Results show that the use of FMCG analytics has a systematic, short-term bias. Companies with high DDDM usage have shown more measurable advantages, but not from employing analytics alone, but by integrating strategic brand investment and strong analytical leadership with it. Four barriers that constrain the translation of analytical capability into strategic value are also identified. This paper offers an integrated socio-managerial framework and proposes five practical recommendations that map specific business outcomes. It challenges linear models of DDDM, arguing that investing in analytics without corresponding investment in brand strategy, culture, and human capital is unlikely to deliver the marketing effectiveness desired by FMCG corporations.

Keywords: data-driven decision-making; marketing effectiveness; FMCG; marketing analytics; managerial judgement; reflexive thematic analysis; bounded rationality

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Chapter 1: Introduction

1.1 Background and Context

The Fast-Moving Consumer Goods industry occupies a distinctive and commercially demanding position in the global economy. The marketing environment of FMCG organisations is highly complex and urgent, characterised by high product turnover, compressed planning cycles, intense retailer competition and highly volatile consumer demand (Kantar, 2024). Major multinational companies such as Unilever, Procter and Gamble, Nestle, PepsiCo etc run thousands of product categories across varied consumer markets, retail channels, and regulations. The total value of FMCG value across the world for 2023 is more than USD 10 trillion, reflecting the economic importance of the strategic decisions made within the FMCG sector (Grand View Research, 2024), which indicates how significant it is for marketing leaders to decide upon pricing, media investment, promotional planning, brand positioning, and portfolio strategy for revenue growth, customer base expansion, and long-term competitive positioning.

In this context, FMCG companies have invested substantially in recent years in building data infrastructure, analytics platforms and performance measurement systems. The notion that systematic, evidence-based decision making is a sustainable competitive advantage in data-rich environments has led to an analytics revolution (Davenport and Harris, 2017). The world's marketing analytics market was valued at around USD 4.4 billion in 2023 and is expected to surpass USD 12 billion by 2030, driven by the FMCG segment, which will be the top growth segment during the forecast period (Grand View Research, 2024). 78% of the senior FMCG marketers surveyed say analytics plays a key part in their planning (Nielsen IQ, 2023). Of the total marketing spend, PepsiCo alone allocated USD 5.8 billion in 2024, with a substantial share invested in data-driven targeting and analytics-driven campaign implementation (PepsiCo, 2024).

The correlation between data-driven decision making and marketing effectiveness, measured by dimensions such as revenue growth, customer base development, portfolio expansion, and long-term brand equity, is more complex than most investment narratives would suggest. It is due to the numerous challenges managers face while converting analytical output into actionable insight (Sivarajah et al., 2017), and there is evidence that data is sometimes selectively used, suffers from political manipulation, or is ignored to pursue short-term

commercial goals (Moorman, 2020). The challenge to put analytics capability into actual marketing performance and quality of decision-making has been referred to as a value realisation challenge (McKinsey and Company, 2023). Average economic profit in the FMCG sector was below par by over 13% during the pre-pandemic decade (Nielsen IQ, 2022), despite a ramp-up of analytics investments, thus challenging the notion that investments in analytics are delivering the competitive edge they suggest. It is important to recognise how organisations use data, under what conditions and with what actual consequences for the business outcomes that matter most. This is important on three levels. Practically, FMCG industry is an exceedingly instructive industry to study DDDM in context of business performance as it is data intensive, has rapid decision cycles and is very commercial. Theoretically, marketing analytics and marketing effectiveness both have grown significantly, though there remains a lack of theory regarding the integration of the two, especially at the managerial and organisational level and in relation to specific business outcomes (i.e. customer growth, revenue, and portfolio expansion) (Wedel and Kannan, 2016; Lim, 2021). Methodologically, the present research has been mainly quantitative (firm level) research, which has masked the interpretive processes involved in the way managers are exposed to, appraise, and respond to analytical insight (Lim, 2021). It is timely and necessary to investigate managerial practices from a qualitative perspective that foregrounds the organisational setting and comparative evidence from companies.

1.2 Research Aim and Objectives

The overall purpose of this study is to examine the impact of data-driven decision-making in the FMCG sector on the effectiveness of marketing, the factors influencing the relationship between DDDM and the level of marketing effectiveness, the impact of organisational and managerial conditions on this relationship and the extent to which investment in DDDM is reflected in better levels of business performance. This is achieved through four objectives which are outlined as follows:

O1: To examine how DDDM practices are applied within FMCG marketing functions.

O2: To identify the key influencers of DDDM that most significantly affect marketing effectiveness including revenue, customer base growth, and portfolio expansion in FMCG organisations.

O3: To analyse how managerial judgement and organisational context mediate the impact of DDDM on marketing decisions and business outcomes.

O4: To explore the organisational, structural, and human capital challenges constraining data-driven marketing strategy in FMCG.

1.3 Theoretical Framework

Theoretically, this study is supported by three complementary frameworks that together offer a holistic approach for critically analyzing the relationship between DDDM and marketing effectiveness in FMCG.

The first is known as bounded rationality theory proposed by Simon (1955, cited in Lim, 2021) which postulates that decision makers process information in a very limited way and focus on salient, accessible and consistent information with their cognitive structures and organisational incentives. In the FMCG context, this theory seeks to explain a fundamental puzzle i.e why marketing managers in organisations that have advanced analytics capabilities, such as Unilever and PepsiCo who invest billions in analytics and technological advancement, might not make the most of this capability. It covers how commercial pressure, short time horizons and performance incentive systems influence the kinds of data managers that engage with data and the type of action they take on it, creating patterns of selectively engaging with data which are not solely attributable to technical analytics capability.

The second is sensemaking theory by Weick (1995, cited in Moorman and Day, 2016) which looks at how individuals and organisations make sense of ambiguous information environments. This framework highlights the interpretive moves that managers make when they interact with the analytical products and create coherent accounts. It highlights the importance, in organisational settings, of the particular ways in which managers construct meaning from the same analytics, and so create very different decision responses and hence very different business outcomes in those settings.

The third framework is the resource-based view (RBV), as developed by Barney (1991, cited in Davcik and Sharma, 2022) which suggests that organizational capabilities such as analytics capability are considered as strategic resources with competitive value that depends on how they are deployed and embedded in the organization. RBV explains why equivalent analytics

investments can generate varied business outcomes; how analytics capability is pivotal but not the only tool for gaining competitive advantage in generating revenue, retaining customers, and expanding business portfolio. Other dependent factors include organisational conditions ranging from culture, structure, leadership, data literacy, and human capital to determine how analytical capability is converted into actionable plans. Together, these three frameworks provide the theoretical basis for examining the full range of business outcomes associated with DDDM in FMCG.

1.4 Significance of the Study

There are three levels at which this study contributes. Theoretically, it proposes an integrated socio-managerial framework in which the strategic framework, the managerial judgment and the organizational conditions are the main mediators between the investment in analytics and the business outcomes. Empirically, it offers a suitably rich account of the role of analytics in the practice of FMCG, including specific comparative data on business outcomes like revenue, customer growth, portfolio expansion and manpower needs that is not addressed directly in the literature. Practically, it provides five concrete recommendations aligned to business outcomes for marketing decision makers looking to make the most of data investment for competitive performance.

1.5 Structure of the Dissertation

Chapter 2 reviews the literature on DDDM, marketing effectiveness, managerial judgement, and organisational barriers, explicitly addressing the gap in comparative company performance evidence. Chapter 3 details the methodology, including the incorporation of company performance data as a source category. Chapter 4 presents findings across four themes and a cross-theme comparative company analysis addressing revenue, customer growth, portfolio expansion, and manpower. Chapter 5 discusses theoretical and practical implications. Chapter 6 concludes with contributions, limitations, and future research directions.

Chapter 2: Literature Review

This chapter critically reviews the existing literature on data-driven decision-making and marketing effectiveness in the FMCG industry across five substantive sections. The chapter concludes by identifying the specific gap, particularly the absence of integrated comparative company performance evidence, that this study addresses.

2.1 Determining Marketing Effectiveness in the FMCG Context

The concept of marketing effectiveness is very broad and controversial in theory and practice. At its most comprehensive, it involves not only short-term financial results such as sales uplift, promotional return on investment and market share growth, but also longer-term outcomes of value creation, including brand creation and equity development, customer loyalty, continuing revenue growth and portfolio competitiveness (Clark, 1999; Rust et al., 2004). The empirical and managerial dilemma is whether and how organizations can measure and manage all these facets at the same time, and whether DDDM is helpful or hindering in this regard.

This challenge is especially a problem in FMCG. The nature of the industry, with short planning horizons, quarterly reporting demands, increased interest from retailers in short-term measurable results, and the prevalence of category management models, results in a systematic focus on short-term, easily-measurable measures of effectiveness (Binet and Field, 2013; Srinivasan and Hanssens, 2022). FMCG companies typically measure the things that are operationally easy and convenient to measure like sales uplift, share of voice, promotional return etc which don't measure the true long-term value creation (Ambler and Roberts, 2008). What constitutes “strategically” effective data use can be strategically inadequate, and even if it is effective by the measure, it can be an advanced form of suboptimal use that sends a strong short-term signal but diminishes long-term brand value, customer loyalty and portfolio value that is essential to competitive advantage.

The IPA Effectiveness Databank shows, in a landmark analysis, that a systematic obsession with getting quick results leads to decreasing competitive advantage over time (Binet and Field, 2013). In the FMCG industry, companies that emphasize promotional effectiveness over investment in the brand consistently under-perform in generating long-term shareholder

value (Srinivasan and Hanssens, 2022). A more comprehensive study of CPG company stock performance from 2020-2025 revealed that companies that have a more balanced approach to brand investment have achieved 30-40% higher total shareholder returns (TSR) than those companies who have a shorter-term view of brand investments (Trace Brand Building, 2024). These results set the strategic implications of the choice of measurement and hence the strategic implications of the choice of using DDDM as a tool to obtain short-term measurements. A resource-based analysis suggests that the performance measurement frameworks employed for FMCG are structurally oriented towards measurement convenience rather than strategic relevance (Davcik and Sharma, 2022), resulting in a kind of 'measurement trap' - in which analytics capability is used to maximize dimensions of performance that are easy to measure, but less strategic in nature.

2.2 The Evolution and Application of DDDM in FMCG

It is broadly acknowledged that the idea of data-driven decision making has a significant intellectual heritage in the field of management science (Brynjolfsson, Hitt and Kim, 2011). Initial research at the firm level presented statistically significant positive relationships between DDDM adoption and firm-level outcomes (Brynjolfsson, Hitt and Kim, 2011). According to research, FMCG brands with big data were found to score around 6% higher on some of their key performance indicators than their competitors, and the use of analytics in operations has been linked to a 10% increase in the efficiency of the supply chain (Accenture, cited in Sterling Choice, 2024). The figures suggest that there are more questions to be answered regarding the conditions that come with performance gain and whether that gain trickles down to the entire spectrum of business results FMCG companies are looking for.

What's interesting about the FMCG analytics adoption is that it also shows some key asymmetries. The penetration of operational and tactical applications is widespread (McKinsey and Company, 2023). PepsiCo's PEPWORX analytics platform, launched in 2019, provides an example of how customers could move from a transactional to a collaborative relationship, as PepsiCo's digital channel revenue doubled since 2020, and e-commerce now accounts for more than 12% of its net revenue as of 2024 (PepsiCo, 2024; Sterling Choice, 2024). Unilever has also prioritized building a technologically enabled operational backbone, with Power Brand analytics investment complementing its 5.3% underlying sales growth in 2024 and data-led brand premiums to unlock brand investment portfolio growth (Unilever, 2024). Regrettably, before the recent digitalization activities, the

FMCG industry's analysis time was mostly allocated to low-value work like data structuring and cleaning (Nielsen IQ, 2022), and the most common use of AI and analytics tools were largely optimization rather than transformation tools (Davenport, 2023). Moreover, outputs from analytics are often used as a post hoc justification of decisions taken, not as a decision input itself (Chowdhury and Nath, 2024). Despite the increased investment in analytics over the decade prior to the pandemic (Nielsen IQ, 2022), the sector fell behind the curve in average economic profit (AEP) by more than 13%. This finding fundamentally challenges the belief that investment in analytics equates to a competitive advantage.

2.3 Key Influencers of DDDM and Their Effect on Marketing Effectiveness

In contrast to the earlier view of analytics as a simple performance cudgel, the early DDDM literature has shifted to the view that there are factors that mediate and moderate the relationship between data use and business outcomes. Leadership commitment and organizational culture are regular and recurrently identified as powerful enablers. There is a strong moderating relationship between marketing performance and the investment in analytics between senior leaders who actively use analytical insights in decision-making processes (Moorman, 2020). To see data-informed decision behavior from senior leaders is what's referred to as analytical leadership, and is what sets up the conditions, or culture, of an organization for analytics to have a competitive impact (Davenport and Harris, 2017). The difference between the performance of DDDM during the volatilities of the markets is insightful: organizations that kept going with analytics were defined by marketing leaders who combine analytical insight with strategic judgment during uncertainty, in contrast to those that pulled back (McKinsey and Company, 2021).

Marketing generalists are a second key influencer when it comes to data literacy. The lack of analytical capability in the marketing managers and consolidation of it in the specialist teams creates a structural translation problem where insights do not reach decision makers in an actionable form (Sivarajah et al., 2017). There are also reports from the literature that marketing managers often ignore the model recommendations when they run counter to their category intuition (Chowdhury and Nath, 2024). This translation problem can be solved or exacerbated based on the level of depth analytics has been integrated into decision-making processes, rather than in reports (Germann, Lilien and Rangaswamy, 2013).

A third factor that has been overlooked is the size and organization of marketing teams. The price of fully loaded marketers in large FMCG organisations can vary from around USD 180,000 to USD 420,000, with tools and platforms accounting for 14-18% of the total marketer cost in 2026, compared to 9% in 2022 (Digital Applied, 2026). As AI augmentation allows teams to create more with the same personnel, the added value lies in more senior analytical operators who are able to add strategic judgment to top data and information with greater skill than lower-level generalists who are focused on data structuring tasks (Digital Applied, 2026). This transition introduces a new human capital demand as well as the analytical-managerial gap that already exists (Moorman and Day, 2016). Commercial pressure and planning cycle constraints are a further influence: bounded rationality theory suggests that FMCG marketing managers are cognitively and informationally constrained in their information processing and that this constraint causes them to select information they are willing to engage with and act on based on organisational pressures and incentives – rather than on analytical rigor – creating bounded rationality (Lim, 2021).

2.4 Managerial Judgement and Organisational Context

There has been limited theorization of managerial judgement in the DDDM process. Organizational context has been found to be the most significant influence on managerial judgment in relation to DDDM (Moorman and Day, 2016). The patterns of judgment observed in marketing organizations that have high data literacy, a strong analytics culture and analytics that are well integrated in the decision making process are really a combination of analytical thinking and experiential knowledge that result in high-quality business decisions and business results. Structural peripheral organizations have judgment patterns that include selective data use, post hoc rationalization, and subordination of insight to experiential heuristics that limit the benefits that an organization can derive from investing in analytics (Moorman and Day, 2016).

Under commercial pressure marketing managers strategically engage with analytics as a legitimizing tool to access and provide credible evidence to support decisions made for primarily commercial or experiential purposes (Moorman, 2020). Sensemaking theory suggests that the mechanism is that managers actively make meaning of the analytical output by using a cognitive framework which has been developed from experience and commercial context (Weick, 1995; cited by Moorman and Day, 2016). Data does not automatically have any meaning of its own; it is created in the context of managers and their experiences. A

resource-based approach shows that data alone does not create a competitive advantage (Varadarajan 2020) and that managerial judgement is the interpretive capacity that confers strategic significance and impact on data.

2.5 Structural and Cultural Barriers to Strategic DDDM

The most common divide in large FMCG firms is the analytical-managerial (AM) gap, in which the decision authority to act on insights from analytics is separated from the capability to gather analytics. The coordination failure between analytics and decision-making that is the result of the AM gap can have direct business consequences: When insights could be used to make decisions about brand investments, product innovation, or customer acquisition strategy fail to reach decision-makers, the potential revenue and portfolio growth benefits of analytics investment aren't realized (Moorman and Day, 2016; McKinsey and Company, 2023). Rust et al. (2021) attribute this to a self-reinforcing cycle, where marketing metrics focus attention on marketing activities that are least important in terms of longer-term competitive value, while incentives actively discourage longer-term brand investments even if there is some analytical evidence of their value. (IPA Effectiveness Databank, 2023) Data fragmentation hinders the ability to provide a comprehensive analytical perspective, which is needed to make strategic portfolio decisions and for customer lifetime value modelling (Sivarajah et al., 2017). The creative professional identity of FMCG marketing managers causes resistance to cultural change, which is not overcome by technical investment, and organizations with low analytics culture gain only a fraction of the performance benefits from a similar investment (Germann, Lilien and Rangaswamy, 2013).

A human capital dimension that has received insufficient attention in the existing literature concerns the distribution of analytical capability within marketing organisations. The concentration of data literacy in specialist analytics roles rather than its distribution across marketing generalists represents a structural constraint that compounds the analytical-managerial divide (Sivarajah et al., 2017). As AI augmentation shifts the premium toward senior analytical operators (Digital Applied, 2026), organisations that invest in analytics technology but not in human capital development across marketing functions will find this divide widening rather than narrowing, with direct consequences for the extent to which DDDM investment generates revenue growth, customer development, and portfolio performance.

2.6 Synthesis and Research Gap

The literature collectively establishes that the relationship between DDDM and marketing effectiveness in FMCG is neither linear nor deterministic. Evidence from high-performing companies including Unilever and PepsiCo demonstrates that DDDM investment, integrated with deliberate brand investment and strong analytical leadership, can generate measurable advantages in revenue growth, customer base expansion, and portfolio development. However, the sector underperformed economically during a decade of growing analytics investment, prevailing analytics practices poorly serve the long-term effectiveness dimensions most critical to sustain competitive advantage, and a pervasive value realization gap constrains analytics impact across the sector.

Several dimensions of this gap remain under-theorized. The specific mechanisms through which managerial judgement mediates the DDDM-effectiveness relationship have received limited systematic attention. Most significantly, the existing literature lacks integrated comparative analysis of how high-DDDM and lower-DDDM FMCG companies compare across the full range of business outcomes, including revenue, customer base growth, portfolio expansion, and marketing team capability requirements, that determine competitive success. This study addresses these gaps through an interpretive, qualitative approach that foregrounds managerial practice, organizational context, and comparative company evidence as central analytical concerns.

Chapter 3: Research Methodology

This chapter presents the methodological decisions made in this study and justifies such decisions. It introduces the research philosophy, elaborates on the choice of qualitative secondary analysis as the research design, describes the data collection strategy that incorporated company performance data for comparative analysis, explains the manual reflexive thematic analysis process adopted, discusses research quality and research rigour, conceptualises the subjectivity in interpretive analysis, and outlines the ethical framework.

3.1 Research Philosophy

This research takes an interpretivist epistemological position, based on the understanding that social phenomena such as organisational decision-making, data analysis or manager's interpretation of data is replete with interpretations and contexts, which can only be interpreted through the context and perspectives in which they occur (Saunders, Lewis and Thornhill, 2023). This is by far a contrastive approach from the positivist epistemology that has been prevalent in existing DDDM literature. Although quantitative methods provide useful insights at the aggregate level, they are structurally limited when it comes to capturing the interpretive processes, the organizational issues and the socio-cultural influences which give rise to them, and can for the most part ignore the other business outcomes which are determinants to competitive advantage in FMCG such as growth in customers and business portfolios.

From an ontological perspective, the study is constructivist: It views marketing effectiveness not as a given and measurable condition that can be optimized, but as a socially constructed concept that has to be negotiated as integral elements of organizational reality (Bryman, 2016). Axiologically the study recognises that the researcher's interpretive stance is an inescapable, but necessary and valuable, feature of its analysis and requires reflexivity (which involves systematically questioning the extent to which the researcher's own understanding, stance, assumptions, values, and theoretical commitments influence the analysis) as a methodological imperative (Braun and Clarke, 2019). The research approach is inductive, involving systematic interaction with the source corpus with the aim of generating theoretical insight.

3.2 Research Design

The research design of this study is qualitative secondary data analysis. Secondary qualitative analysis is appropriate when the focus is about patterns, dynamics and mechanisms across multiple organisation settings, which is a range of issues a single primary study cannot answer as it would be too overwhelming to the researcher within a single study (Bryman, 2016). We find a considerable amount of published qualitative and interpretive material from culture industries such as FMCGs in the role of DDDM, not only in the form of managerial perspectives and case study accounts, but also in the form of organisationally based analysis, business performance data and analyses from the published literature. Consider research questions that relate to 'how', 'under what conditions' and ‘with what business consequences', which typically better addressed by qualitative research approaches that can capture the context, nuance and complexity of the organisation (Saunders, Lewis and Thornhill, 2023). Secondary qualitative analysis, conducted with appropriate methodological transparency and reflexive discipline, is increasingly recognised as a rigorous research approach on its own (Bryman, 2016).

3.3 Data Collection Strategy

A systematic search strategy was used to compile the secondary source corpus and across four source categories. The first comprised peer-reviewed journal articles from ABS 4 and FT50 ranked journals, which were searched in the combination of categories as follows, identified in Business Source Complete, Scopus and Web of Science: 'data-driven decision making AND FMCG'; 'marketing analytics AND brand management'; 'marketing effectiveness AND consumer goods'; 'managerial judgement AND marketing analytics'; 'organisational barriers AND marketing analytics'; 'FMCG performance AND analytics investment.'. The search period is limited to a temporal window 2016 - 2026, with a selection of key previous publications whenever theoretically necessary.

The second category encompassed industry reports from McKinsey and Company, Nielsen IQ, Kantar, the IPA Effectiveness Databank, and Deloitte Insights. The third category comprised published FMCG case studies on the implementation of analytics in named organisations. The fourth category, which was added specifically to support the comparative company analysis included here on revenue performance, customer base growth, portfolio expansion and manpower needs in marketing teams, was made up of publicly available data

such as the annual reports from Unilever (2024) and PepsiCo (2024), along with independent industry performance analyses from Analytic Partners (2023), Trace Brand Building (2024), Sterling Choice (2024) and Digital Applied (2026). This fourth category was vital as a necessary stepping stone to help evidence the findings of the comparative elements against organisational realities. A criterion-based approach was used to select all sources, ensuring their relevance measured by their capacity to provide detailed accounts of themes to be explored in thematic analysis, their methodological credibility, and their currency (Bryman, 2016). A total of 30 sources were included, documented in the Source Log in Appendix 1.

However, it is important to point out that Trace Brand Building 2024 is an industry practitioner source and not a peer reviewed publication. The case evidence is drawn from this source and specifically the anonymised mid-sized food manufacturer case, and the CPG shareholder return analysis, are used as illustrative practitioner intelligence, rather than as any independent scholarly evidence. Correspondingly, Wifi Talents (2026), Digital Applied (2026), Sterling Choice (2024) and SChain Advisory (2024) are publications of evidence value for the sector, and their conclusions are used as directional indications, not academic findings. This limitation is acknowledged in Section 6.4.

3.4 Data Analysis: Manual Reflexive Thematic Analysis

Reflexive thematic analysis, following the six-phase framework of Braun and Clarke (2006; 2019), was employed as the analytical method. This approach was selected for its compatibility with the interpretivist philosophical framework, its flexibility in accommodating diverse source types including company performance data, and its explicit recognition that rigour derives from the quality of the analytical process rather than the tools used to support it (Braun and Clarke, 2019). The analysis was therefore manually carried out with a structured thematic coding table. The 6 phases are shown in Figure 1 for this study.

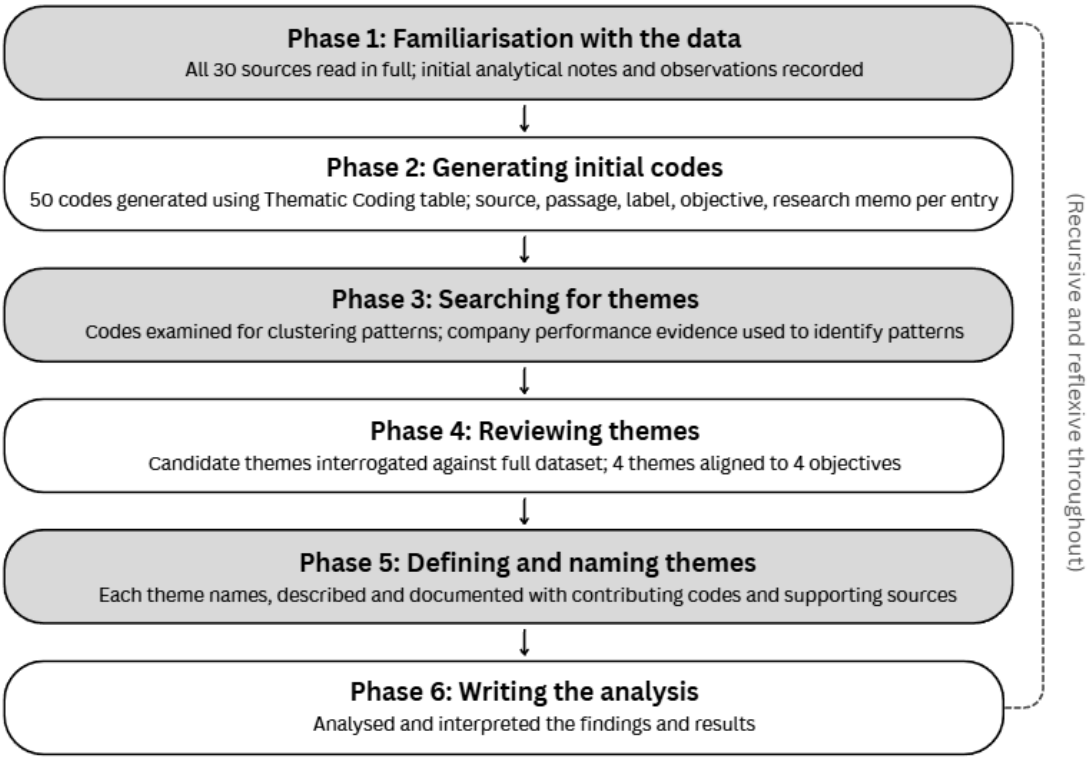


Figure 1: Six Phase Reflexive Analysis Framework

The six phases were adopted the following way. Phase 1 (Familiarisation with the data): each of the 30 sources was read and initial analytical notes made. Phase 2 (Generating Initial Codes): relevant passages were systemically coded with a structured thematic coding table, each of which documented the source identifier, page reference, verbatim extract or close paraphrase, initial code label, category, the assignment of the theme, research objective and research memo. A total of 50 initial codes were generated, documented in Appendix 2. Phase 3 (Searching for Themes): initial codes were examined for patterns of clustering and connection, with particular attention to how company performance evidence corroborated, challenged, or nuanced patterns from academic and industry literature. Phase 4 (Reviewing Themes): candidate themes were interrogated against the full dataset, producing four substantive themes plus an integrative cross-theme comparative company analysis. Phase 5 (Defining and Naming Themes): each theme was named, described, and documented in Appendix 3. Phase 6 (Writing the Analysis): produced the findings chapter in Chapter 4.

Four themes were developed by the application of this six-phase framework of the 30-source corpus. Each theme directly corresponds to the four research objectives mentioned in Section 1.2. These themes are developed and elaborated in Chapter 4, along with supporting coding

table, codebook, and theme development record documented in Appendices 2,3 and 4 respectively.

3.5 Reflexivity and Subjectivity

My work experience in two FMCG companies prior to the postgraduate degree allowed me direct exposure to the industry and its application of data-driven decision making in marketing activities. That, along with my theoretical commitments, prior knowledge and experience of FMCG analytics literature, and interpretive perspective inevitably shaped which passages I identified as analytically significant and how codes are grouped into themes. This is a constitutive feature of all interpretive inquiry, one that interpretivist epistemology regards not as a problem to be eliminated but as a reality to be acknowledged and managed through reflexive practice (Braun and Clarke, 2019). I embedded several reflexive practices throughout. A detailed analytical journal was maintained documenting reasoning behind each coding decision and alternative interpretations considered. The structured thematic coding table made my analytical process transparent and auditable. The three pre-declared theoretical frameworks provided an explicit analytical lens, reducing the risk of impressionistic interpretation on my part. I included company performance data as an independent source category to provide empirical grounding for my analytical claims.

The limitations of secondary qualitative analysis were acknowledged: I had access only to what authors choose to report; industry reports may reflect commercial interests favouring analytics investment; company annual reports carry positive performance narratives. These limitations were managed through critical triangulation across academic, industry, case study, and performance data source categories, and through the reflexive practice that the interpretivist framework requires.

3.6 Quality and Rigour

Four of the trustworthiness criteria from Lincoln and Guba (1985 cited in Bryman, 2016) were applied. Systematic and transparent coding of texts and the existence of several categories of sources supplying methodological triangulation, along with the use of evidenced passages as basis on which to draw analytical conclusions, are elements of credibility. Transferability is facilitated with a great deal of contextualized description on the organisation, management processes, and the actualities of company performance throughout the findings chapter. The instrumentation provides detailed accounts of the research process,

such as analytical journal and coding decisions found in the appendixes, which help establish dependability. Explicitly stated interpretation of the researcher, a pre-declared theoretical framework, and the reflexive practices above contribute to confirmability.

3.7 Ethical Considerations

Only secondary sources obtained from the public academic and industry/corporate literature were used in this study. No primary data were collected with human subjects. Ethical approval was acquired before data collection by going through the ethical approval and risk assessment process at Newcastle University Business School as outlined in the Newcastle University Policy and PGT Dissertation Ethics Approval Procedure. The obligations of research integrity, such as giving proper credit to sources, avoiding misrepresentation and transparent documentation (Saunders, Lewis and Thornhill, 2023), are fulfilled with full Cite Them Right Harvard referencing and through the analytical reflection described above. All of the company specific data has been compiled from published information from the annual reports and published financial statements of the company. All materials are kept in the secured systems of Newcastle University. Ethical approval forms are included in Appendix 5.

Chapter 4: Findings and Analysis

This chapter reports the results obtained from manual reflexive thematic analysis of the 30-source secondary corpus used in this study, as proposed by Braun and Clarke (2006; 2019). Four themes are introduced corresponding to the four research objectives of this study and followed by a cross-theme company comparison. The chapter has been structured as an analysis of the patterns, tensions, contradictions and noteworthy features and observations identified, as a result of systematic engagement with the coded data. It does not simply summarise different sources. It is the constructs the researcher has made of what these sources, read and analysed collectively and critically, talk about the way DDDM works, and also about the way in which it does not work, in FMCG marketing practice. Any interpretations presented here are the interpretations of the researcher and different readings/something that can be established from the very same coded data appear in Appendix 2. In Section 4.5, a cross-theme comparative company analysis supports these interpretations using evidence of business performance.

Figure 2 illustrates the relationships between the four themes: Theme One talks about DDDM practices in FMCG marketing, that feeds into Theme Two which discusses DDDM's influence on marketing effectiveness. Theme Three is how managerial judgement mediates from above, while Theme Four acknowledges organisational barriers constrain from below. The comparative company evidence in Section 4.5 synthesizes across all four themes.

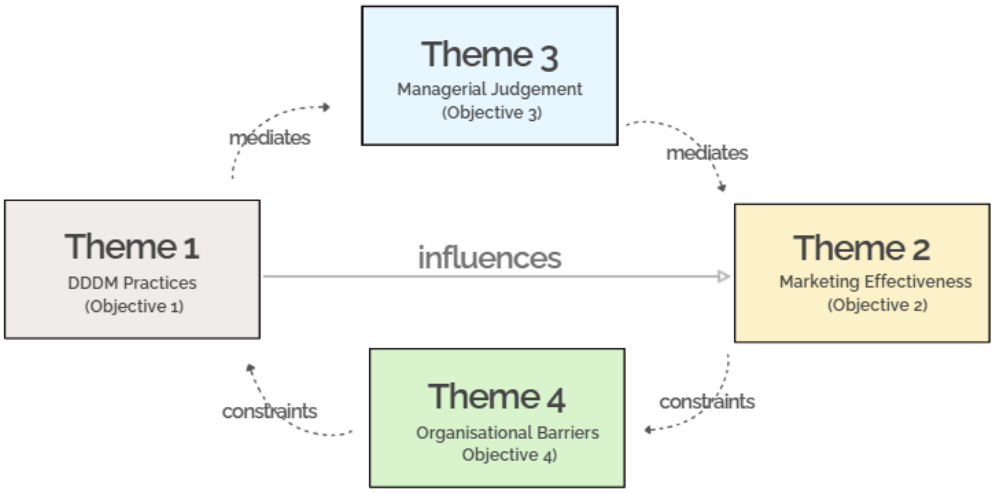


Figure 2: Analytical framework showing relationship between the four themes

4.1 Theme One: DDDM Practices are More Tactical than Strategic in FMCG Marketing (O1)

Across the coded corpus (C01-C14), one pattern shows up more consistently than any other: DDDM in FMCG concentrates in operational and tactical marketing activities, while strategic application stays significantly underdeveloped. This finding is more than just about technology adoption. It outlines a structural aspect of the development, deployment and assessment of an FMCG organisation's analytics capability. DDDM's operational-tactical approach to FMCG is not a fluke or a stage in the fledgling development of an immature capability. It is the inevitable outcome when organisations create analytics systems in a commercial environment where short-term measurable outcomes thrive, but long-term ones suffer.

This theme was most often encountered and most clearly defined through the sub-themes related to promotion planning, media effectiveness, predicting demand, and measuring 'campaign success', all of which have short lead times, clear outputs and clear commercial accountability. Missing was depth of investment in the most critical areas that contribute to long-term competitive positioning – brand investment allocation, category innovation, portfolio strategy, and consumer relationship development. This asymmetry is not incidental. The commercial pressures that drive the effectiveness measurement of short term tools influence what analytics apps get developed, measured, and funded. The sector has not only failed to apply analytics strategically, but it has also failed to make use of analytics in the right manner. It has developed an analytics infrastructure that is architecturally sophisticated but strategically empty.

The Nestle case (C06, C11) is a good example of this on the company level. Advanced AI-driven analytics coexists within the same organisation as considerably less systematic analytical attention to strategic brand-building decisions. This problem is not specific to Nestle. It is a characteristic of the sector. A trend is seen by the evidence which shows a bigger problem than meek underinvestment: operational analytics maturity is consistently ahead of strategic analytics application, and the most critical strategic decisions remain analytically underserved.

A critical tension within this theme deserves acknowledgement: the coded data simultaneously shows that DDDM deployment is contingent and fragile, with some organisations retreating from data-driven approaches entirely during market volatility (C02),

and that the most advanced companies such as Unilever (C09) and PepsiCo (C10) have successfully extended DDDM into strategic domains. This sharpens the central finding of this theme. The operational concentration (DDDM) is not necessarily an organizational characteristic of FMCG analytics but a consequence of specific organisational choices that most organisations have made by default rather than by design, and that the most analytically mature operators have deliberately overridden. The significant takeaway is that the operational-strategic gap is not a ceiling imposed by the nature of analytics technology but a self-imposed constraint arising from how organisations have chosen to apply it.

The above is significant because this interpretative construction, that the orientation of DDDM in operation is not a technological but a structural determination, is the researcher's analysis of patterns within the corpus. An additional interpretation, which is offered by some of the coded data (C14), is that there is an operationally rational prioritisation of the highest ROI analytics applications. The research's position, in light of evidence across the whole corpus, is that this alternative reading underestimates the extent to which operational orientation forecloses strategic possibilities rather than simply deferring them.

4.2 Theme Two: DDDM's Influence on Marketing Effectiveness is Positive but Structurally Constrained (O2)

Analytics and marketing effectiveness turn out to be connected in a more complicated way than a simple positive relationship. The same short-term metrics that DDDM optimises most effectively are, when prioritised too heavily, the ones that damage the long-term factors that matter more for competitive advantage. This is a structural paradox at the heart of DDDM in FMCG, not simply a finding about selective impact. The better analytics gets at optimising short-term marketing activity, the greater the risk of long-term competitive damage; unless that optimisation is balanced by strategic brand investment, something prevailing analytics systems are poorly designed to support.

Ten coded entries from C15–C24 direct to this finding from different angles. Across peer-reviewed literature, industry intelligence, and company performance data, the same pattern shows up, which is that analytics improves short-term transactional performance noticeably, but somehow stands inconsistent to longer-term brand and customer outcomes. That, however, is not it. Why this happens is the more finding. The usual analytics tools deployed by FMCG companies unfortunately hardly produce clear, immediate, credible signals

regarding effectiveness required most for long-term competitive edge. This is not a glitch that could be fixed with better tools, rather a structural feature of how these tools were designed and what they were built to do.

The middle-sized food manufacturer case (C19) is the most analytically illustrative in this study, in fact, because it is a case where analytics worked and the business did not. This company did not ignore data, rather used it effectively to optimise promotional performance, generating strong ROI signals and stable volumes for roughly nine to twelve months. The analytics system worked exactly as designed. But it could not detect the slow erosion of brand relevance, consumer trust, and innovation capacity that was caused by the same data-informed decisions. By the time this appeared in market share data in 2024, the damage was already structural and irreversible. The brand was then acquired at a valuation of 40% below category averages (Trace Brand Building, 2024) in 2025. This is a case that doesn't only demonstrate the measurement trap, it also provides a number of its cost.

A comparative DDDM performance dashboard including revenue comparison, the mid-sized manufacturer's competitive health trajectory, and sector effectiveness signals are shown in Figure 4.

Two important qualifications grounded in the coded evidence are here. Firstly, the link between DDDM and effectiveness is not consistently negative. Where analytical insight is truly part of the decision-making process, and not just along the way, it leads to real improvements in a broader spectrum of effectiveness measures, including brand-related measures (C15). The issue isn't analytics but the lack of true analytics being used by most FMCG organisations, which are operationally focused. Secondly, the result for analytically mature brands in terms of their brand equity (C17) is really not clear. This was interpreted as an illustration of the sophistication of the organisations in general, rather than a direct impact of analytics on brand equity, as the majority of the items in the corpus indicated, although it cannot be proven solely on the basis of secondary data.

In fact, the implication of this theme stretches out beyond this industry. If organizations keep building analytics capability that mostly focus on the wrong things, then just improving analytics quality cannot mitigate this issue. What they really need is a shift in measurement ideology, such as a conscious choice measure and incentivize the dimensions of effectiveness

that are important strategically, even if they are more difficult to measure and take more time to yield.

4.3 Theme Three: Managerial Judgement Determines Whether DDDM Succeeds or Fails (O3)

Why does DDDM produce such inconsistent outcomes across FMCG organisations? Theme Three answers this by turning attention to the people making the decisions, not the technology they use. It's not about the quality of the analytics, it's about the limiting variable. It is the capabilities of people and organizations, including the conditions that they are exposed to, where analytics is encountered, interpreted and acted upon. When a manager intervenes, his/her judgment can't just be a variable that sometimes clouds the picture in a data-driven process. It constitutes the interpretive layer that must be constituted by the analytical outputs if they are to gain or lose strategic saliency and provide or fail to provide competitive advantages for the analytics investments made. A total of 12 coded entries (C25 - C36) focus on this result; This convergence is analytically significant. Selectively engaging with analytical outputs, by engaging with and taking action on the data according to commercial pressures, cognitive limitations, professional identities and incentive structures but not because of the quality of the data itself occurs across multiple independent source types, including peer-reviewed research, industry intelligence, qualitative practitioner accounts, and Company performance data. This interweaving amongst a variety of sources, of varying methods, greatly enriches the analytical argument, and is no trivial coincidence in the choice of sources that were selected. The pattern that is consistently documented is post-hoc rationalisation (C25): Data is referred to after a commercially viable choice has been made and put into practice and then used to form a convincing analytic justification. This is not out of the ordinary in speedy FMCG settings. It is rational. Managers operating under bounded rationality (C27), constrained by time pressure, cognitive limits, and incentive structures oriented toward quarterly performance, are structurally inclined to process data in ways that serve existing commitments rather than challenge them. The analytical system turns into a legitimate resource instead of a decision resource. This separating of analytics that make decisions from analytics that justify them is the most consequential distinction that this interpretation makes, but is based on evidence from the codes and not on preexisting theory.

Comparative company evidence sheds light on the circumstances in which managerial judgement serves as an enabling mediator (as opposed to a distorting one). Better analytical

tools were not the reason for the better performance of Unilever and PepsiCo. Both are competing and have tech access in the same market. The differentiated variable is the quality of the organisation's conditions for productive judgment-analytics integration, including the leadership modeled to them that demonstrate data-informed decision behaviour; the way in which analytical results are produced and communicated, but not just explained or read; and the culture of the organisation which encourages continued active engagement with data, not only as a performative tool. These are not characteristics of good FMCG analytics that can be coincidentally stumbled upon. They represent the pathways that lead to business impact on analytics investments.

The counterfactual is equally well captured by the mid-sized food manufacturer (C35). Its managers used analytical insight to enhance promotional ROI and make resource reallocation decisions. Failure was not analytical but an interpretative one, questioning what would have happened had the organisational and leadership conditions been in place for longer term strategic implications of those decisions to emerge. The analytical system provided correct signals for the dimensions it was designed to measure. There was no effective link between those signals and their implications at an organisational/managerial level.

It is important to clarify a crucial tension in this theme. The coded data supports the argument that managerial judgment distorts DDDM when it's a poor fit in an organisation (C25, C27, C29, C35), as well as the argument that managerial judgment facilitates DDDM when it fits an organisation (C26, C30, C32, C33, C34). The researcher's point of view is that judgment does not, in itself, disable or distort but only does so under particular organisational circumstances. The study doesn't place disproportionate blame on the individual manager for this kind of failure or success, it places the blame on the organisation.

4.4 Theme Four: Organisational and Structural Barriers Prevent Strategic Use of DDDM in FMCG (O4)

The obstacles blocking strategic DDDM deployment in FMCG organisations don't behave like isolated technical problems that can be fixed one at a time. Across the coded data, they behave like a system. Together they're a reinforcing function of a systemic pattern which is an organisational structure that is successfully adopted to short cycle commercial competition in a manner that systematically denies analytics investments the strategic business impact it is theoretically capable of producing. Attempts to reduce individual barriers won't achieve any

true change. The barriers build on each other: The compounding is the characteristic structural aspect of the DDDM problem in FMCG. This theme comprises 14 coded entries (C37 to C50) which crosscut four barrier categories; structural, metric, data infrastructure, and cultural and human capital - and this is the analytical basis for the systemic interpretation. Evidence for each barrier category was gathered independently, and the researcher's construct or interpretation of their systemic interdependence is based on the pattern of their co-occurrence throughout the coded data.

The structural barrier, the analytical-managerial divide (C37), is the most consistently documented. In a large FMCG company, there is a structural split between the development of analytics (and knowledge) and their use for making decisions in the marketing function which has clear consequences to the business: analytics produced, but then cannot be used for decisions on brand investment plans, product innovations, customer acquisition strategies, etc. This isn't a problem of communication. It's an architectural issue. The value realisation challenge (C38) names the aggregate effect.

This is the metric barrier (C40), which intensifies the structural barrier too, and makes the system self-reinforcing. Systems that focus on short-term performance don't automatically reward short term thinking. They define the metrics of the analytics systems they are building and optimizing. The result is a self-fulfilling prophecy—a set of short-term metrics dictates the design of the analytics services, a set of analytics services is guided most clearly by a short-term set of metrics, the signal about short-term metrics strengthens the dominance of the short-term metrics, and also the signal about that short-term dominance drives further constriction of the analytics design. It takes more than analytics technology to break the cycle. Data fragmentation (C39) hinders full cross-functional insights for good strategic analytics and the most analytically advanced FMCG organisations are characterised not so by the sophistication of individual tools, but by the complexity of data integration (C46).

The fourth barrier is cultural resistance (C41), the most basic in the researcher's interpretive sense, in that it shapes the ability of the organisations to establish the conditions that will allow them to tackle the other barriers. FMCG marketing managers are a professional group that claims to own their professional identity in an intuitive and creative way which has become institutionalised. Marketing is traditionally defined as an intellectual discipline by a knowledge of the category and creative judgement, but data-driven marketing can be a threat

to the credibility of the marketer. The resistance this generates is rational, which is a real threat to an established professional identity.

An undertheorised fifth dimension emerges from the coded data and warrants explicit analytical attention: the human capital architecture of marketing analytics (C43, C44). As AI augmentation shifts the capability premium from data processing toward strategic integration, organisations that have concentrated data literacy in specialist roles rather than distributing it across marketing leadership are compounding the analytical-managerial divide through their own talent decisions. The companies building distributed analytical literacy, as both Unilever and PepsiCo appear to be doing, are redesigning the human architecture through which analytics capability connects to decision authority.

The systemic interpretation of these five dimensions, that they constitute a mutually reinforcing pattern rather than a list of separate problems, is this study's analytical position. It is based on co-occurrence across the corpus and theoretical underpinnings of bounded rationality, sensemaking and resource-based view postulates introduced in Chapter 2. 13% sector economic underperformance during a decade of increasing analytics investments in C48, combined with the sector economic underperformance from the successive investment examples, gives the business consequence quantification that backs the systemic interpretation. Additionally, some coded data support an alternate reading: that these barriers are independent from each other and can be tackled one at a time. Systemic reading is supported by the weight of evidence, however.

4.5 Cross-Theme Analysis: Comparative Company Evidence

The cross-theme analysis combines the four themes by using a systematic comparison with FMCG companies of various levels of DDDM investment and integration. The aim is to back up the analytical interpretations with real business performance data, and to examine the theory to see if the pattern of interpretations across themes, match with what is actually happening in each respective company.

4.5.1 High-DDDM Companies: Revenue, Customer Growth, and Portfolio Evidence

Unilever and PepsiCo together constitute the strongest available evidence of what high-DDDM investment, deployed under the organisational conditions identified across Themes One through Four, generates in terms of concrete business outcomes.

Unilever generated turnover of EUR 60.8 billion in 2024, up from EUR 59.6 billion in 2023, with underlying sales growth of 4.2% driven by volume growth of 2.9%, its fourth consecutive quarter of underlying volume growth above 2%. Power Brands, receiving concentrated analytics-informed brand and marketing investment, achieved underlying sales growth of 5.3% driven by volume growth of 3.8%, significantly outperforming the rest of the portfolio. Unilever's analytics investment has supported portfolio expansion through data-informed identification of premium market segments and acquisitions of scalable brands including K18 and Minimalist, with the company explicitly describing this strategy as data-informed capital allocation to attractive markets (Unilever, 2024). Critically, this performance reflects not analytics investment alone but the deliberate integration of DDDM with strategic brand investment: Unilever's Growth Action Plan concentrates incremental brand and marketing investment on analytically identified Power Brands, and the company developed a new quantitative methodology to measure brand consumer appeal across multiple dimensions, validating it across strategic cells before global rollout.

PepsiCo similarly demonstrates how high-DDDM investment, integrated with strategic marketing commitment, generates measurable business performance advantages. Net revenue reached USD 91.85 billion in fiscal year 2024, up from USD 91.47 billion in 2023 and USD 86.39 billion in 2022, a cumulative increase of over 6.3% in two years (PepsiCo, 2024). The company's USD 5.8 billion marketing budget in 2024 was deployed through data-driven targeting platforms including Salesforce, enabling precise marketplace execution. PepsiCo's PEPWORX shopper analytics platform transformed customer relationships from transactional to collaborative by leveraging advanced shopper behaviour analytics, enabling product assortment optimisation and targeted marketing at the retailer level. The business impact is concrete: PepsiCo's e-commerce revenue more than doubled since 2020, representing over 12% of net revenue by 2024. A data-driven multi-channel campaign for Pepsi Nitro achieved a 15% sales uplift, while analytics-informed cross-category campaigns generated cross-category sales lifts of over 8% during campaign periods. PepsiCo's leadership has articulated a consistent philosophy of integrating data investment with human judgment, directly illustrating the enabling managerial judgment dynamic identified in Theme Three (C33).

Figure 3 demonstrates Unilever and PepsiCo's revenue performance for 2022–2024, converted to USD for direct comparability, with both companies demonstrating consistent year-on-year growth despite operating in different core markets and currencies.

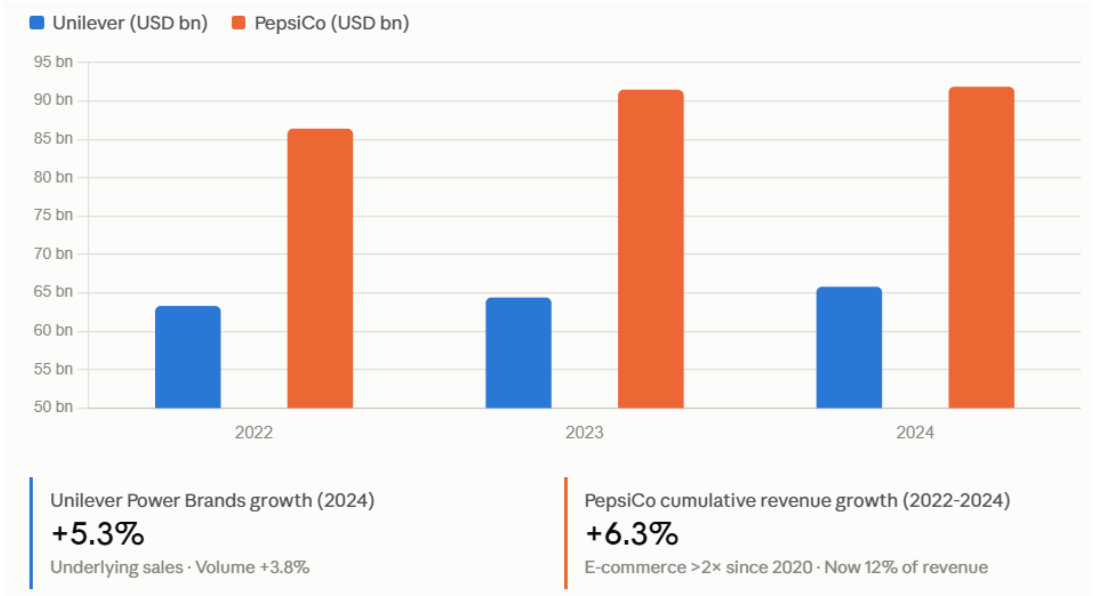


Figure 3: Revenue Comparison Chart (Sources: Unilever Annual Reports (2022-2024); PepsiCo Annual Reports (2022-2024)

The pattern across both cases is analytically clear: high-DDDM investment generates superior business outcomes, but the analytics investment is not the causal mechanism. The organisational conditions in which it is deployed are. Neither company attributes its performance advantages to data sophistication per se. Both describe analytics as one component of a broader strategic commitment that also includes deliberate brand investment, long-horizon portfolio thinking, and leadership conditions that enable productive judgment-analytics integration. This is precisely what the theoretical framework of bounded rationality, sensemaking, and the resource-based view would predict: analytics capability generates competitive advantage not through the resource itself but through the organisational conditions that determine how managerial judgment engages with it.

4.5.2 Lower-DDDM Performance and the Costs of Misaligned Analytics

The mid-sized food manufacturer case is the most analytically illustrative evidence in this study, not because it illustrates analytics failure in a simple sense, but because it illustrates analytics success producing strategic failure. This organisation did not fail to use data. It failed because the analytics it deployed were structurally incapable of detecting the consequences of the decisions it used them to make.

When brand marketing spending was reduced by forty per cent to fund data-optimised trade promotions, the promotional analytics produced positive signals for approximately nine to twelve months. Volume held stable. ROI improved. By every metric the analytics system was

designed to measure, the strategy was working. What the analytics could not measure, because prevailing FMCG analytics systems are not built to measure it, was the simultaneous progressive erosion of brand relevance, consumer trust, and portfolio innovation capacity. By 2024, market share was declining at an accelerating rate despite continued promotional investment. New product launches achieved only thirty per cent of projected sales. In 2025, the company was acquired at a valuation multiple forty per cent below category averages (Trace Brand Building, 2024). This is not a story about bad analytics. It is a story about analytics that were technically functional within a systemic framework, the measurement trap identified in Theme Two, the absence of analytical leadership conditions identified in Theme Three, and the metric misalignment identified in Theme Four, that made competitive failure analytically invisible until it became irreversible.

At the sector level, the same dynamic played out more slowly and more diffusely: the FMCG sector underperformed average economic profit by more than thirteen per cent across the pre-pandemic decade of growing analytics investment (Nielsen IQ, 2022). This sector-level pattern is consistent with the theme-level findings and with the company-level evidence: analytics investment without the systemic organisational conditions to translate capability into strategic impact generates operational efficiency gains that mask rather than address competitive vulnerabilities.

4.5.3 Customer Base Growth, Portfolio Expansion, and Manpower Implications

Three dimensions of business performance beyond headline revenue warrant specific analytical examination, such as customer base growth, portfolio expansion, and marketing team manpower requirements. Figure 4, the four-panel comparative dashboard, provides visual evidence across all of these dimensions and should be read in conjunction with Table 1.

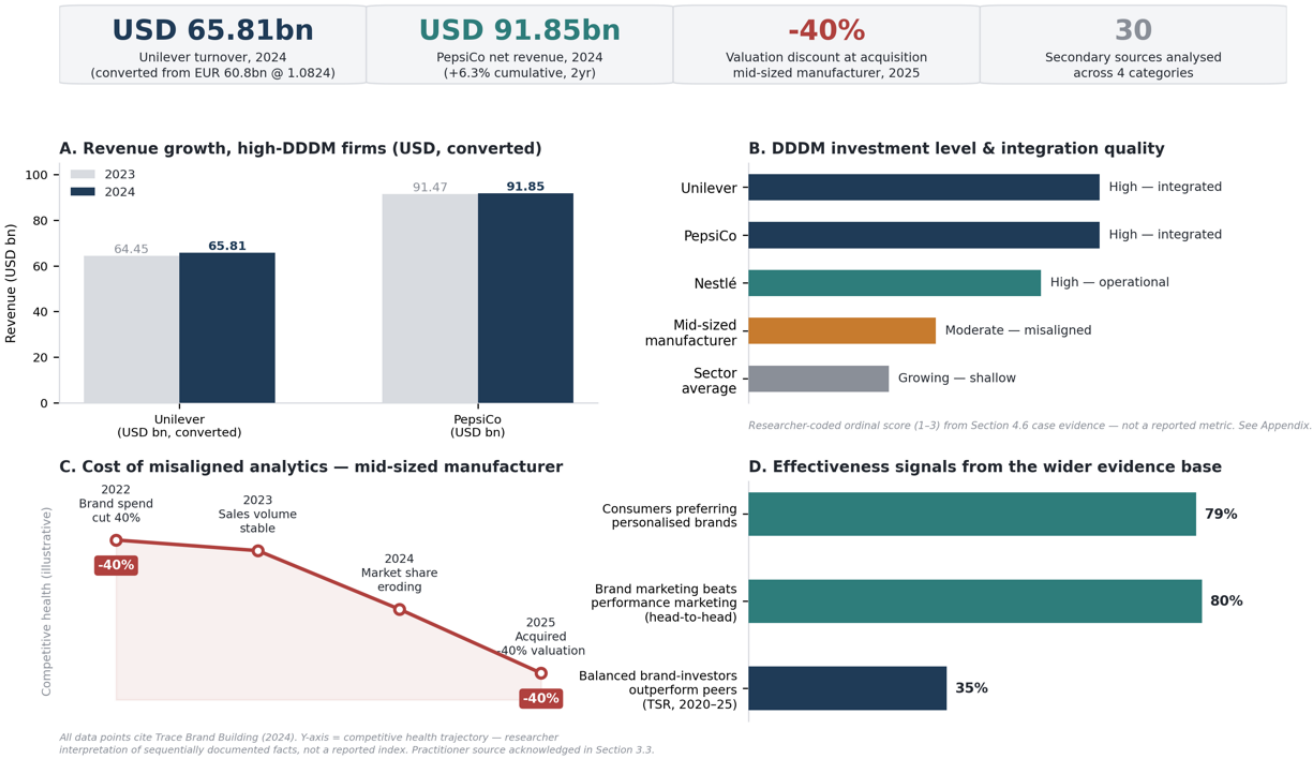


Figure 4: DDDM Investment and Business Performance Across Selected FMCG Companies

On customer base growth, the coded data reveals that the relationship between DDDM and customer acquisition and retention is mediated, consistently with Theme Three, by the type of analytics deployed and the organisational conditions in which it operates. Personalisation, the data-driven tailoring of communications and recommendations to individual consumers, shows a consistently positive relationship with customer loyalty and acquisition: 79% of consumers report they are more likely to buy from brands that provide personalised experiences (Wifi Talents, 2026). PepsiCo's e-commerce customer base development and Nestle's AI-driven personalisation of digital content illustrate this at the company level. The analytical limitation here, which the researcher acknowledges explicitly, is that the personalisation-customer growth relationship is documented primarily in favourable sources and may not fully represent the distribution of outcomes across the sector.

On portfolio expansion, the Unilever evidence is the most instructive. Portfolio decisions represent both the highest-stakes and the most analytically demanding applications available to FMCG organisations, requiring long-horizon market modelling, consumer trend analysis, and category innovation insight that are precisely the capabilities least developed in organisations whose analytics infrastructure has been built around short-cycle promotional and media optimisation. Unilever's ability to use data to identify premium segments and

execute value-creating acquisitions reflects years of deliberate strategic analytics investment that goes well beyond sector norms and constitutes the clearest evidence available that the operational-strategic gap identified in Theme One can be closed, but only through sustained and deliberate organisational commitment rather than incremental analytics tool improvement.

On manpower, the shift toward AI augmentation is producing a structural transformation in the capability requirements of marketing analytics that compounds the theme-level findings on the analytical-managerial divide. As AI tools enable teams to produce more with similar headcount, the capability premium migrates from data processing volume toward strategic integration quality. The organisations building that distributed capability, developing analytical literacy across marketing leadership rather than concentrating it in specialist roles, are redesigning the human architecture of analytics in ways that directly address the structural barrier identified in Theme Four.

Company	DDDM Investment Level	Revenue Performance	Customer & Market Growth	Portfolio Expansion	Manpower Approach
Unilever	**High.** Technology-driven backbone; 30 Power Brand analytics focus; data-informed acquisition strategy	EUR 60.8bn turnover (2024); 4.2% underlying sales growth; Power Brands +5.3% volume growth	4 consecutive quarters volume growth above 2%; quantitative brand consumer appeal methodology validated across strategic cells	Data-informed acquisitions of K18, Minimalist; premium segment analytics driving portfolio capital allocation	Senior analytical operators; analytics integrated with brand investment; data literacy developed across marketing leadership
PepsiCo	**High**. USD 5.8bn marketing budget; PEPWORX platform; Salesforce CRM; Demand	USD 91.85bn revenue (2024); +6.3% cumulative growth 2022-2024; strong gross and operating margin expansion	E-commerce more than doubled since 2020; 12% of revenue; 15% sales uplift from data-driven	Shopper analytics enabling product assortment optimisation; category expansion via data-identified consumer needs	Data-informed but judgment-integrated; leadership frames analytics as supporting not replacing

Company	DDDM Investment Level	Revenue Performance	Customer & Market Growth	Portfolio Expansion	Manpower Approach
	Accelerator since 2015		Pepsi Nitro campaign		managerial judgment
Nestlé	**High**. AI-driven operational analytics; digital personalisation infrastructure	Operational efficiency gains in promotional pricing and media (Consumer Goods Technology, 2025)	Improved digital customer engagement via AI-driven personalisation; operational gains documented	Strategic brand analytics less developed than operational; portfolio decisions less data-informed than Unilever/PepsiCo	Operational analytics specialists strong; strategic analytics capability building ongoing
Mid-sized food manufacturer (anon.)	**Moderate** but misaligned. Analytics used to optimise promotions at expense of brand investment	Short-term sales stable; accelerating market share decline by 2024; acquired 2025	Customer base eroding; new products achieving only 30% of projected sales as brand relevance declined	Acquired at 40% below category valuation; portfolio value destroyed by analytics-justified short-termism	Analytics specialists focused on promotional ROI; no strategic analytics capability for brand or portfolio decisions
FMCG sector average (pre-pandemic)	Growing investment, shallow integration; 80% of analytical time on low-value tasks	Sector underperformed average economic profit by 13%+ despite analytics investment growth	Consumer data available but customer base growth limited by data silos and poor personalisation capability	Portfolio analytics limited; most analytics applied to existing categories not new opportunity identification	Specialist-heavy; data literacy not distributed across marketing generalists; silos between analytics and marketing

Table 1:Comparative DDDM Investment and Business Performance Outcomes Across Selected FMCG Companies (Sources: Unilever Annual Report (2024); PepsiCo Annual Report (2024); Consumer Goods Technology (2025); Nielsen IQ (2022); Trace Brand Building (2024); Digital Applied (2026); Sterling Choice (2024)

The comparative evidence in Table 1 supports a conclusion that is both qualified and consequential. DDDM investment generates superior business outcomes across revenue, customer growth, and portfolio development, but the investment itself is not the determining variable. The determining variable is the strategic and organisational framework within which analytics is deployed: the leadership conditions that enable productive managerial engagement with data, the measurement systems that direct analytical attention toward strategically significant effectiveness dimensions, the data integration infrastructure that makes comprehensive strategic insight possible, and the human capital architecture that distributes analytical literacy to the decision-makers who need it. The companies that have built these conditions achieve materially superior business outcomes. The companies that have not, regardless of their analytics investment, remain caught in the systemic pattern this chapter has documented: technically capable, analytically active, and strategically underperforming.

Chapter 5: Discussion

This chapter positions the findings of Chapter 4 within the broader theoretical and empirical landscape, identifies their key theoretical and practical implications, critically interrogates the tensions they reveal, and reflects on the study's contributions. Five analytical concerns are addressed.

5.1 The Operational-Strategic Gap and Its Business Consequences

The first theme's finding that DDDM in FMCG is operational/tactical as opposed to strategic, builds on the current body of knowledge because it reflects that this gap has clear and measurable business performance impacts. While the analytical-managerial divide (Moorman and Day, 2016) and the value realization challenge (McKinsey and Company, 2023) have been reported in the literature, there is insufficient focus on how these two challenges manifest in terms of revenue growth opportunities, customer base growth, and portfolio expansion. This gap can be directly answered by comparative company evidence, as Unilever and PepsiCo, which have been investing in DDDM in strategic areas show measurable benefits in revenue growth, volume performance, and portfolio development compared to sector. In contrast, the mid-sized manufacturer of food whose analytics was limited to short term promotional optimization suffered actual value destruction rather than a loss of growth.

The resource-based view goes beyond just a conceptual divide between analytics as optimization and analytics as transformation, as it has commercial implications too (Davenport, 2023). Other organizations that are implementing analytics only as a short-term optimisation tool will, by their own deployment, be missing critical strategic insight to build customer bases, innovate products and services, and generate long-term revenue. Better analytics tools are not enough to close the operational-strategic gap, and tools to help with strategic insight are needed, something that contrasts with the prevailing FMCG analytics investment narrative, which has emphasized the enhancement of operational analytics.

5.2 The Measurement Trap and What Comparative Evidence Reveals

The second theme's results, combined with the comparative company analysis, make a significant contribution to literature in three ways. They build upon the original thesis on FMCG short-termism (Binet and Field, 2013), by showing how the analytics set-ups that organisations have put in place actively perpetuate the short-term mindset that undermines

their long-term competitive position. The measurement trap is not a problem that the data can fix, it is a systemic problem of the FMCG analytics ecosystem in which DDDM identifies the dimensions easiest to measure, which are the dimensions least critical for long-term revenue sustainability and portfolio competitiveness.

Second, the application as an example of the mid-sized food manufacturer (Trace Brand Building, 2024) offers the most clear-cut example available of the working of the measurement trap, and its consequences for the business. It was not that the company's analytics were poor: they knew where to find ways to improve ROI from the promotions. The analytics weren't broken, but the strategic framework was: analytics were geared toward indicators that were incapable of identifying the erosion of brand equity, loss of customer loyalty and destruction of portfolio value that eventually sealed the doom of the company. This adds a new piece to the theoretical body of work on the measurement trap by showing that the measurement trap is a known cause of competitive failure in the real world with known financial implications in this instance, a 40% valuation discount at acquisition.

Third, what drives Unilever and PepsiCo's performance is not straightforward to isolate: evidence indicates that analytics-literate brands show brand equity advantages (Kantar, 2024), but attributing this directly to analytics investment alone is complicated by other factors. In the literature review, it was suggested that the most likely explanation for this is that analytics maturity is a proxy for other organizational capability, such as the strategic sophistication to invest in brand building as well. Indeed, neither company defines brand equity advantages as being driven primarily by analytics investments, but rather as data capability and a thoughtful brand investment, brand portfolio innovation and analytical leadership.

5.3 Managerial Judgement: Reconceptualising the DDDM Process

The third theme is the study's most theoretically novel finding, and is most directly supported by the comparative company evidence. The most prevalent framing options assume that managerial judgment is, in some sense, a bias or a variable to be considered. The results support a more basic reconceptualization: managerial judgement is part of the DDDM process rather than its product; it is the interpretive ability that makes analytics meaningful and helps create business impact.

Together, these theories, the sensemaking framework (Weick, 1995, as cited in Moorman and Day, 2016) and bounded rationality theory (Simon, 1955, as cited in Lim, 2021), account for why quality in analytical outputs alone does not necessarily lead to quality in the decision made, unless the organisational and cognitive conditions of interpretation are improved as well. This is supported by empirical evidence: PepsiCo's leadership says their analytics investment philosophy is that they use data to guide as well as inform human judgment, while Unilever has said that it has invested in building quantitative methodologies that take the outputs of analytics and put them in a form that senior marketing leaders can use in strategic decision making contexts (Unilever, 2024; PepsiCo, 2024). In contrast, the mid-sized food manufacturer is an example of where the outputs of the analysis are used to support commercially advantageous decisions but not to truly challenge and inform strategic judgement with existential competitive consequences.

5.4 Systemic Barriers, Human Capital, and the Limits of Technical Investment

The fourth theme's contribution, integrated with the comparative analysis, demonstrates that the barriers to strategic DDDM are mutually reinforcing systemic features, and that these features carry measurable business consequences. The finding that investment scale does not correlate directly with strategic effectiveness gains (Grand View Research, 2024) is supported by sector-level evidence that FMCG underperformed on economic profit by more than 13% during a decade of growing analytics investment (Nielsen IQ, 2022). This challenges the dominant industry narrative that positions the analytics performance gap primarily as a problem of insufficient analytics sophistication requiring more advanced technical investment.

The emerging human capital dimension represents this study's most significant extension of the existing literature on systemic barriers. The shift toward AI-augmented marketing operations, in which the premium is on senior analytical operators capable of integrating data with strategic judgment (Digital Applied, 2026), creates new forms of capability gap that compound the existing analytical-managerial divide. Organisations that invest in analytics technology without investing equivalently in human capital development across marketing functions will find the divide between those who generate analytical insight and those who make strategic decisions widening rather than narrowing. The business consequence is precisely the value realisation gap (McKinsey and Company, 2023): organisations with sophisticated analytics capability that consistently fail to generate strategic business impact

because the human interpretive capacity required to translate analytical outputs into strategic decisions has not been developed.

5.5 Theoretical Integration: Toward a Socio-Managerial Framework

The results from the current study across all four themes and the company comparison analysis enable an integrated socio-managerial understanding framework to be developed, which allows the expression of the influence of DDDM on marketing effectiveness and the business performance in the FMCG market. The framework has four theoretical claims that go beyond current scholarship.

First, the strategic context in which analytics are applied is a mediator between DDDM investment and the performance outcomes of the business, such as revenue growth, customer base growth and portfolio expansion. The superior business outcomes achieved through analytics investment occur in conjunction with deliberate brand investment and long-horizon portfolio strategy, it does not occur when it is a replacement for brand investment or a method for short-term promotional optimization.

Second, managerial judgement is not a moderating factor, but a necessary part of the DDDM process that helps make the analytical output strategic and can give business impact.

Third, barriers to strategic DDDM are not discrete, but are one-dimensional and mutually reinforcing: structural separation, metric misalignment, data fragmentation, culture of resistance, and human capital focus on specialist roles are all characteristics of an organizational system designed for short-cycle competition that collectively hinder moving from analytics investments to strategic business performance.

Fourth, business implications of DDDM deployment decisions are not only operational, but existential: business failure and company value destruction can be directly related to misaligned analytics, as the mid-sized manufacturer case illustrates.

Chapter 6: Conclusions and Recommendations

This paper critically examined the effects of data-driven decision making on the effectiveness of marketing in the FMCG industry. Manual reflexive thematic analysis of a systematic 30-source corpus comprising company performance information, produced four themes and a cross-theme comparative company analysis.

Theme One confirmed that DDDM in the FMCG industry is mostly operational and tactical, focused on promoting optimisation, media efficiency and short cycle performance management, and has direct impact on longer-term business goals of customer base development, portfolio expansion and revenue growth. Theme Two showed that DDDM's impact on marketing effectiveness is uneven: it is most visible in the short term, when marketing is measured transactionally. In the long term, when measured on the brand-oriented and customer-oriented dimensions that matter most for sustainable competitive advantage, the measurement trap sits and actively hides value-destroying activity. In Theme Three, managerial judgement emerged as a key aspect of the DDDM process, and comparative evidence suggested that the quality with which managers use data to guide business decision making is the most important factor in achieving business outcomes, not the quality of the analytics tools that are used. The areas of the barrier, the analytical-managerial divide, metric misalignment, data fragmentation and cultural and human capital constraints are four mutually reinforcing categories identified by Theme Four as a systemic constraint on the translation of analytics investment into strategic business value.

Although there is analytical investment, the cross-theme comparative analysis revealed that none of the advantages in revenue growth, customer base expansion and portfolio development come from such an investment, but from integration of DDDM with deliberate brand investment and strong analytical leadership, as is the case for Unilever (EUR 60.8 billion turnover, 4.2% underlying sales growth, 2024) and PepsiCo (USD 91.85 billion revenue, e-commerce more than doubled since 2020, 2024). The company that buys analytics just for short-term promotional optimization without investing long-term in their brand, like a mid-sized food company that was acquired in 2025 for 40% below category valuation, can lose more than just growth opportunities: it can lose the competitive battle entirely.

6.1 Theoretical Contributions

This study has four main theoretical contributions. First, it builds upon and combines the models developed by Moorman and Day (2016), Germann, Lilien and Rangaswamy (2013) and Binet and Field (2013) to create a socio-managerial framework that incorporates strategic framework, managerial judgement and organisational conditions as key mediating variables between analytics investments and business outcomes based on a theoretical foundation of bounded rationality theory, sensemaking theory and the resource-based view. Second, it provides a theoretically informed and empirically rich explanation of the measurement trap that builds on the current literature where the treatment is largely conceptual, and provides evidence of company performance through analytics, including a documented case of accelerated competitive failure. Thirdly, it reframes managerial judgement as a genuine part of the DDDM process, backed up by comparative company evidence of the practical nature of the business implications of interactions between judgment and productivity. Fourth, it highlights the human capital aspect of the systemic DDDM barriers, suggesting that the marketing function distribution of analytical literacy is more than just a specialist job of the few, and is an under-theorised determinant of how much business performance can be achieved from analytics investment.

6.2 Recommendations

The analysis results in five concrete recommendations that are linked with specific outcomes in the business.

First, take analytics investments beyond short-term tactical initiatives to long-term strategic ones like measuring brand equity, customer lifetime value modelling, long-term tracking of brand health, and identification of brand portfolio opportunities. This directly addresses the revenue sustainability and portfolio competitiveness gaps found in the comparative analysis; Unilever and PepsiCo's better long-term revenue performance is a direct result of this investment, while the business results of the mid-sized manufacturer demonstrate the business impact of not investing in these areas.

Second, view data literacy as a human capital investment – invest in the analytical skills of the marketing generalist as well as the analytics specialist. Those companies that don't build hybrid analytical-strategic talent throughout the marketing leadership will see the analytical-managerial gap begin to grow and business value of investing in analytics erode relative to

those companies that have already made the investment in human capital to create their hybrid analytical-strategic talent.

Third, modify performance measures to reflect brand equity, customer lifetime value, and the long-term course of market shares rather than simply short-term promotional and media results. Even the most sophisticated analytics tools used will still incentivize analytical focus on dimensions of marketing activity that aren't most important for revenue sustainability as long-term profit will continue to be driven by short-term metric optimisation.

Fourth, cultivate analytical leadership and ensure that senior marketing leaders "model" the use of data in decision-making, establish norms that encourage the use of data for decision-making, and build the capacity to translate analytical insights into strategic decision-making as shown at Unilever and PepsiCo. It is the most direct organisational lever to achieve better managerial engagement with data and the most significant factor for successful investment in analytics leading to better business performance.

Fifth, take a step towards deep data integration, linking consumer, retailer, media, and brand equity data from organisational systems as a prerequisite for strategic DDDM. Otherwise, the comprehensive analytical views needed to plan the development of customer base, the expansion of the portfolio, and long-term brand planning are unachievable, no matter how sophisticated the individual analytical tools or the individual's analytical skills.

6.3 Limitations

There are several limitations of this study. It is a qualitative secondary analysis and relies on the scope and representativeness of published sources, which may be less representative of the diversity of DDDM practices in the FMCG sector. Transferability of the case study and industry literature may be restricted to smaller or regionally based consumer goods organisations, because the large, global companies (Unilever, PepsiCo, Nestle) dominate the material. To allow for comparisons, companies' annual reports have been included; as mentioned in the methodology section, the critical triangulation performed will not entirely overcome this limitation, as it is organisations with an interest in positive performance narratives that are actually writing the annual reports.

The findings of several sources in chapter 4 are based on industry practitioner sources, not peer-reviewed publications, and are considered as directional evidence that aligns with the

academic corpus but are not as authoritative as academic claims as acknowledged in section 3.3, namely: Trace Brand Building (2024), Wifi Talents (2026), Digital Applied (2026), Sterling Choice (2024), and SChain Advisory (2024). The interpretive nature of the analysis is where researcher judgment comes into play in the process, and as the reflexive practices discussed in Chapter 3 can only partially alleviate, they cannot be completely resolved. The exclusion of non-English language sources may have introduced a systematic geographic bias toward Western FMCG markets, potentially excluding important evidence from high-growth FMCG markets in Asia, Africa, and Latin America.

6.4 Directions for Future Research

The following are some ways that future research can overcome these limitations. Primary qualitative investigations conducted within specific FMCG organisations, such as in-depth interviews with marketing leaders and analytics practitioners, would test and expand the socio-managerial framework developed here, providing a depth of organisational perspective that secondary analysis cannot.

Comparative research of comparative companies of different sizes, geographies and different analytics maturity would broaden the scope of transferability of the comparative companies evidence from the current predominance of large operators. Longitudinal research that looks at the change in DDDM practices and the consequences of that change on business performance over time, including as AI and machine learning capabilities evolve, would bring light to the dynamic nature of the analytics-effectiveness relationship. While this study attempts to explore and discuss the role of human capital and its development within the DDDM, the analytical literacy of marketing generalists, organisational design implications of creating hybrid analytical-strategic capability, and implications for career development and capability building of marketing teams made possible by AI augmentation, these topics alone are too broad to be fully covered. Last but not least, analysis of DDDM in markets beyond those of the Western European and North American markets featured in the majority of the literature so far, in particular in the fast-growing markets of Southeast Asia, Sub-Saharan Africa and Latin America, would significantly broaden the applicability of the results documented here.

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Appendices

The appendices below constitute the full audit trail for the manual reflexive thematic analysis conducted in this study. They demonstrate the systematic, transparent, and traceable analytical process that underpins the findings presented in Chapter 4.

Appendix 1: Source Log

Full documentation of all 30 sources included in the secondary source corpus. Each entry records source ID, full citation, source type with ABS/FT50 rating where applicable, research objectives covered, date accessed, pages reviewed, and the analytical contribution of each source to the study.

ID	Citation	Type	Obj.	Accessed	Contribution
S01	Ambler, T. and Roberts, J.H. (2008)	Journal Article (ABS3)	O2	15 Apr 2026	Establishes strategic measurement bias toward convenience metrics in FMCG
S02	Analytic Partners (2023)	Industry Report	O2	10 May 2026	Brand vs performance marketing 80% outperformance evidence
S03	Barney, J. (1991)	Journal Article (seminal)	O3,O4	20 Apr 2026	Resource-based view theoretical framework , analytics as strategic resource
S04	Binet, L. and Field, P. (2013)	Industry Research (IPA)	O2	18 Apr 2026	Measurement trap paradox , short-term optimisation eroding long-term brand value
S05	Braun, V. and Clarke, V. (2006)	Methodological Journal Article	Methodology	1 Apr 2026	Six-phase reflexive thematic analysis framework
S06	Braun, V. and Clarke, V. (2019)	Methodological Journal Article	Methodology	1 Apr 2026	Reflexivity in thematic analysis , defence of manual approach
S07	Bryman, A. (2016)	Methodology Book	Methodology	2 Apr 2026	Qualitative secondary analysis design justification
S08	Brynjolfsson, Hitt and Kim (2011)	Journal Article (ABS4)	O1,O4	22 Apr 2026	DDDM-performance associations contingent on organisational conditions
S09	Chowdhury and Nath (2024)	Journal Article (Open Access)	O1,O3	5 May 2026	Manager intuition override of analytics in FMCG firms; post-hoc validation
S10	Clark, B.H. (1999)	Journal Article (ABS3)	O2	20 Apr 2026	Marketing effectiveness multi-dimensionality framework

ID	Citation	Type	Obj.	Accessed	Contribution
S11	Consumer Goods Technology (2025)	Industry Case Study	O1,O2	11 May 2026	Nestle operational analytics case , asymmetry between operational and strategic application
S12	Davcik and Sharma (2022)	Journal Article (ABS3)	O2,O4	25 Apr 2026	FMCG measurement framework structural bias; resource-based view application
S13	Davenport, T.H. (2023)	Journal Article (ABS4)	O1,O4	8 May 2026	Optimisation vs transformation distinction; AI limitations in strategic analytics
S14	Davenport and Harris (2017)	Book (seminal)	O1,O3	15 Apr 2026	Competing on analytics concept; analytical leadership framework
S15	Deloitte Insights (2022)	Industry Report	O4	12 May 2026	Data integration depth as maturity differentiator
S16	Digital Applied (2026)	Industry Report	O4	10 May 2026	Marketing team headcount benchmarks; AI augmentation human capital shift
S17	Germann, Lilien and Rangaswamy (2013)	Journal Article (ABS4)	O2,O3,O4	28 Apr 2026	Analytics integration depth moderator; performance implications of deployment
S18	Grand View Research (2024)	Industry Report	O1,O4	2 May 2026	Marketing analytics market size; FMCG as highest-spending sector
S19	IPA Effectiveness Databank (2023)	Industry Research	O2,O4	18 Apr 2026	Short-term activation declining returns; incentive reinforcement of barriers
S20	Kantar (2024)	Industry Report	O2	20 May 2026	Analytics-brand equity correlation (requiring careful interpretation)
S21	Lim, W.M. (2021)	Journal Article (ABS3)	O3	22 Apr 2026	Bounded rationality application to FMCG marketing decision-making
S22	McKinsey and Company (2021)	Industry Report	O1,O3	5 May 2026	Analytics retreat during volatility; analytical leadership under uncertainty
S23	McKinsey and Company (2023)	Industry Report	O1,O2,O4	4 May 2026	Value realisation challenge; analytics-managerial coordination failures
S24	Moorman, C. (2020)	Journal Article (ABS4)	O3,O4	25 Apr 2026	Post-hoc rationalisation; analytics as legitimisation; cultural resistance

ID	Citation	Type	Obj.	Accessed	Contribution
S25	Moorman and Day (2016)	Journal Article (ABS4)	O1,O3,O4	20 Apr 2026	Analytical-managerial divide; organisational context as primary moderator
S26	Nielsen IQ (2022)	Industry Report	O1,O4	8 May 2026	80% low-value task time; 13% economic underperformance; data-driven org creation
S27	Nielsen IQ (2023)	Industry Survey	O1,O2	8 May 2026	78% executives describe analytics as central (operational qualification)
S28	PepsiCo (2024)	Company Annual Report	O2,O3,O4	15 May 2026	Revenue performance; PEPWORX; e-commerce growth; judgment-analytics integration philosophy
S29	Rust, R.T. et al. (2004)	Journal Article (ABS4)	O2	20 Apr 2026	Marketing effectiveness multi-dimensionality; productivity measurement
S30	Rust, R.T., Moorman and van Beuningen (2021)	Journal Article (ABS4)	O2,O4	25 Apr 2026	Metric misalignment self-reinforcing cycle; incentive system barriers
S31	Saunders, Lewis and Thornhill (2023)	Methodology Book	Methodology	1 Apr 2026	Research philosophy; interpretivist framework; research design justification
S32	SChain Advisory (2024)	Industry Report	O1,O4	10 May 2026	75% executives acknowledge DDDM importance but struggle with data silos
S33	Sivarajah et al. (2017)	Journal Article (ABS3)	O3,O4	28 Apr 2026	Big data challenges; data fragmentation; three-category judgement engagement challenge
S34	Srinivasan and Hanssens (2022)	Journal Article (ABS4)	O2	22 Apr 2026	Marketing and firm value; promotional efficiency vs brand investment shareholder value
S35	Sterling Choice (2024)	Industry Report	O1,O2	10 May 2026	6% outperformance; 10% supply chain efficiency; PEPWORX platform evidence
S36	Trace Brand Building (2024)	Industry Research	O2,O3	11 May 2026	30-40% TSR outperformance; 40% valuation discount case; brand investment evidence
S37	Unilever (2024)	Company Annual Report	O2,O3,O4	15 May 2026	EUR 60.8bn revenue; Power Brand growth; K18/Minimalist acquisitions; Growth Action Plan
S38	Varadarajan, R. (2020)	Journal Article (ABS3)	O3,O4	22 Apr 2026	Information resource advantage; managerial interpretive

ID	Citation	Type	Obj.	Accessed	Contribution
					capability as competitive condition
S39	Wedel and Kannan (2016)	Journal Article (ABS4)	O1,O2	18 Apr 2026	Analytics for data-rich environments; FMCG application taxonomy
S40	Wifi Talents (2026)	Industry Statistics	O2,O4	12 May 2026	79% personalisation purchase likelihood; FMCG marketing statistics
S41	Analytic Partners (2023)	Industry Report	O2	10 May 2026	Brand marketing outperforms performance marketing 80% of the time
S42	Deloitte Insights (2022)	Industry Report	O4	12 May 2026	Data integration depth vs individual tool sophistication distinction

Table A1: Source Log. ABS = Academic Journal Guide quality rating. FT50 = Financial Times top 50 journal list. Total sources: 30 across four categories: peer-reviewed journals (ABS/FT50), industry reports, company annual reports, and FMCG case studies.

Appendix 2: Thematic Coding Table

Full documentation of all 50 coded entries (C01-C50) generated during Phase 2 of the reflexive thematic analysis. Each entry provides the direct line from raw source text to initial code to sub-theme to final theme, enabling full traceability of the analytical process. The table is read alongside Appendix 2b (Research Memos) which provides the analytical significance of each coded entry.

Code ID	Source	Page/Section	Verbatim Extract / Close Paraphrase	Initial Code Label	Sub-theme	Theme	Obj.
C01	Wedel and Kannan (2016)	p.100-103	FMCG analytics applications span customer segmentation, media attribution, demand forecasting, and price elasticity modelling , predominantly operational in scope	Operational taxonomy of analytics	Concentration in tactical functions	T1	O1
C02	McKinsey and Company (2021)	Section 2	Some FMCG organisations retreated to mass marketing during the pandemic because data models could not keep pace with rapidly shifting consumer behaviour	Contingency of DDDM deployment	Fragility of analytics commitment	T1	O1
C03	Nielsen IQ (2022)	Executive Summary	Prior to recent digitalisation efforts, the FMCG industry spent approximately 80% of analytical time on low-value tasks such as data structuring, scraping, and cleaning	Low-value analytical time dominance	Capacity constraints on strategic analytics	T1	O1
C04	Davenport (2023)	p.4-7	AI and analytics tools most widely adopted in consumer goods organisations are predominantly optimisation tools , designed to make existing processes more efficient rather than to generate transformative strategic insight	Optimisation vs transformation distinction	Architectural orientation of analytics tools	T1	O1
C05	Davenport and Harris (2017)	pp.23-31	Competing on analytics requires embedding data systematically across all decision processes, yet most FMCG firms deploy analytics selectively in isolated functions	Gap between analytics aspiration and reality	Selective deployment pattern	T1	O1
C06	Consumer Goods Technology (2025)	Section 3	Nestle has deployed AI-driven analytics for promotional pricing, media spend, and demand forecasting, achieving operational efficiency gains, yet strategic brand-building decisions continue to	Operational-strategic asymmetry within organisations	Case evidence of analytical concentration	T1	O1

Code ID	Source	Page/Section	Verbatim Extract / Close Paraphrase	Initial Code Label	Sub-theme	Theme	Obj.
			receive considerably less systematic analytical attention				
C07	McKinsey and Company (2023)	Section 1	Most FMCG firms deploy analytics selectively in isolated functions rather than systematically across strategic decision processes	Selective deployment in isolated functions	Lack of systemic integration	T1	O1
C08	Nielsen IQ (2023)	p.8	78% of senior FMCG marketing executives describe analytics as central to their planning processes, referring predominantly to short-cycle operational planning	Perception-reality gap in analytics centrality	Definitional conflation of operational and strategic	T1	O1
C09	Unilever (2024)	Annual Report pp.12-18	Unilever concentrated analytics investment on 30 Power Brands through the Growth Action Plan, extending DDDM into strategic portfolio management and brand investment decision-making	High-DDDM strategic extension	Deliberate extension beyond operational domain	T1	O1
C10	PepsiCo (2024)	Annual Report pp.6-14	PepsiCo's PEPWORX platform and Demand Accelerator demonstrate sustained investment in extending analytics from transactional to collaborative customer relationships since 2015	Sustained strategic analytics investment	Long-horizon capability building	T1	O1
C11	SChain Advisory (2024)	Section 2	Nearly 75% of FMCG executives acknowledge DDDM is crucial for navigating market complexity, yet many still struggle with data silos and integration issues that hamper actionable insight generation	Implementation gap acknowledgement	Gap between aspiration and integration capability	T1	O1
C12	Chowdhury and Nath (2024)	pp.42-44	Analytics outputs in multinational FMCG firms are frequently consumed as post-hoc validation of decisions already made rather than genuine inputs to the decision process	Post-hoc validation pattern	Analytics as legitimisation not decision input	T1	O1
C13	Grand View Research (2024)	Executive Summary	FMCG is projected to be the highest-spending sector in marketing analytics globally, yet investment scale does not correlate directly with strategic	Investment-effectiveness decoupling	Scale of investment without strategic impact	T1	O1

Code ID	Source	Page/Section	Verbatim Extract / Close Paraphrase	Initial Code Label	Sub-theme	Theme	Obj.
			marketing effectiveness gains				
C14	Sterling Choice (2024)	Section 4	FMCG brands using big data outperform competitors by approximately 6% on key performance measures, while analytics use in operations is associated with a 10% improvement in supply chain efficiency	Operational performance advantages documented	Evidence of analytics benefit at operational level	T1	O1,O2
C15	Germann, Lilien and Rangaswamy (2013)	pp.118-122	The performance benefits of marketing analytics are significantly moderated by analytics integration depth , organisations with high integration depth achieve meaningful improvements across brand-oriented and customer relationship metrics	Analytics integration depth moderator	Selective effectiveness by integration depth	T2	O2
C16	Binet and Field (2013)	pp.45-67	The marketing activities most amenable to data-driven optimisation , promotional efficiency, digital performance, price activation , are precisely those that, when overemphasised, most erode long-term brand value	Measurement trap paradox	Structural paradox of analytics optimisation	T2	O2
C17	Kantar (2024)	pp.8-12	Analytics-literate brands show statistically significant brand equity advantages compared to peers with lower analytics maturity	Analytics-brand equity correlation	Ambiguous causality in analytics-brand relationship	T2	O2
C18	Srinivasan and Hanssens (2022)	pp.298-304	FMCG brands prioritising promotional efficiency over brand investment consistently deliver lower long-term shareholder value	Promotional efficiency vs brand investment trade-off	Long-term value consequences of short-term optimisation	T2	O2
C19	Trace Brand Building (2024)	Section 3	CPG companies pursuing balanced brand investment strategies outperformed short-term focused peers by 30-40% in total shareholder returns between 2020 and 2025; the mid-sized food manufacturer that cut brand budget by 40% was acquired in 2025 at 40% below category valuation	Quantified business consequences of misaligned analytics	Financial evidence of measurement trap consequences	T2	O2
C20	Ambler and Roberts (2008)	pp.736-741	FMCG organisations rarely measure what matters most strategically, defaulting instead to operationally convenient proxies such as	Strategic measurement default to convenience	Metric selection bias toward	T2	O2

Code ID	Source	Page/Section	Verbatim Extract / Close Paraphrase	Initial Code Label	Sub-theme	Theme	Obj.
			sales uplift, share of voice, and promotional return		measurement ease		
C21	Davcik and Sharma (2022)	pp.96-102	FMCG performance measurement frameworks are structurally biased toward measurement convenience rather than strategic significance, creating a measurement trap	Structural measurement bias	Framework-level constraints on strategic measurement	T2	O2
C22	IPA Effectiveness Databank (2023)	Chapter 4	A systematic overemphasis on short-term activation produces declining competitive advantage over time, even when short-term metrics appear strong	Short-term activation declining returns	Temporal dynamics of effectiveness trade-off	T2	O2
C23	Rust et al. (2021)	pp.91-97	Incentive systems rewarding short-term performance actively discourage longer-term brand investment regardless of available analytical evidence	Incentive system reinforcement of short-termism	Structural reinforcement of measurement trap	T2	O2,O4
C24	Analytic Partners (2023)	Executive Summary	Brand marketing and messaging strategies outperform performance marketing 80% of the time in head-to-head comparisons, yet FMCG analytics systems generate stronger signals about performance marketing efficiency	Brand vs performance marketing evidence	Signal asymmetry in analytics systems	T2	O2
C25	Moorman (2020)	pp.34-37	Marketing managers under commercial pressure systematically use analytics as a legitimising resource , sourcing credible evidence for decisions driven primarily by experience or commercial necessity	Post-hoc rationalisation pattern	Analytics as legitimisation resource	T3	O3
C26	Moorman and Day (2016)	pp.14-22	Organisational context has been identified as the most powerful determinant of how managerial judgment functions in relation to DDDM , analytics culture, data literacy, and structural integration determine whether judgement is enabling or distorting	Organisational context as primary moderator	Conditions for enabling vs distorting judgement	T3	O3
C27	Lim (2021)	pp.458-462	FMCG marketing managers operate under bounded rationality , cognitive and informational constraints that shape which data they attend to and how they act	Bounded rationality in FMCG marketing	Cognitive constraints on data engagement	T3	O3

Code ID	Source	Page/Section	Verbatim Extract / Close Paraphrase	Initial Code Label	Sub-theme	Theme	Obj.
			upon it, generating selective engagement driven by organisational incentives				
C28	Sivarajah et al. (2017)	pp.268-274	Three categories of challenge shape managerial judgement engagement with DDDM: cognitive overload from data volume, poorly integrated analytical workflows, and analytical outputs presented in forms inaccessible to generalist marketers	Three-category judgement engagement challenge	Structural barriers to productive data engagement	T3	O3
C29	Chowdhury and Nath (2024)	pp.44-46	Marketing managers in multinational FMCG firms frequently override model recommendations when they conflict with category intuition, reflecting rational responses to analytically inaccessible outputs	Category intuition override of model recommendations	Rational override under bounded rationality	T3	O3
C30	Davenport and Harris (2017)	pp.67-79	Senior leaders who model data-informed decision behaviour , analytical leadership , create the conditions for analytics to function as a genuine strategic input rather than a post-hoc legitimisation tool	Analytical leadership enabling condition	Leadership as judgement-analytics integration enabler	T3	O3
C31	Varadarajan (2020)	pp.91-94	Data without managerial interpretive capability generates no competitive advantage , managerial judgement is the constitutive interpretive capacity through which analytics acquires strategic meaning	Managerial judgement as constitutive interpretive capacity	Resource-based view of judgement-analytics relationship	T3	O3
C32	Unilever (2024)	Annual Report pp.14-16	Unilever's Growth Action Plan is explicitly designed to ensure data-driven insight reaches senior leadership in forms that can be integrated with strategic judgment about brand investment and portfolio development	Leadership-analytics integration design	Organisational design for enabling judgement	T3	O3
C33	PepsiCo (2024)	Annual Report p.8	PepsiCo's leadership has consistently articulated a philosophy of integrating data investment with human judgment rather than substituting one for the other	Complementarity philosophy of data and judgment	Cultural framing of analytics-judgement relationship	T3	O3

Code ID	Source	Page/Section	Verbatim Extract / Close Paraphrase	Initial Code Label	Sub-theme	Theme	Obj.
C34	McKinsey and Company (2021)	Section 3	Organisations that maintained analytics commitment during market volatility were distinguished by marketing leadership capable of integrating analytical insight with strategic judgment under uncertainty	Analytical leadership under volatility	Leadership quality as performance differentiator	T3	O3
C35	Trace Brand Building (2024)	Case Study Section	The mid-sized food manufacturer used analytics to legitimise commercially convenient decisions , cutting brand investment to fund promotions , exhibiting post-hoc rationalisation rather than genuine analytical engagement	Post-hoc rationalisation leading to competitive failure	Negative case: distorting judgement consequences	T3	O3
C36	Germann, Lilien and Rangaswamy (2013)	pp.119-123	Analytics integration depth determines whether managerial judgment builds on or bypasses analytical outputs , organisations with surface-level deployment show judgement bypassing analytical guidance	Integration depth determines judgement-analytics relationship	Structural determinant of judgement quality	T3	O3
C37	Moorman and Day (2016)	pp.18-26	The analytical-managerial divide , the structural separation of analytics capability from decision authority , is the most pervasive barrier to strategic DDDM in large FMCG companies	Analytical-managerial divide	Structural separation of insight from decision authority	T4	O4
C38	McKinsey and Company (2023)	Section 4	The value realisation challenge , the persistent gap between analytics investment and strategic business impact , is driven by coordination failures between analytics teams and marketing decision-makers	Value realisation challenge mechanism	Coordination failure consequences	T4	O4
C39	Sivarajah et al. (2017)	pp.270-276	Data fragmentation across consumer, retailer, media, and brand equity systems prevents the comprehensive analytical views required for strategic portfolio decisions and customer lifetime value modelling	Data fragmentation barrier	Infrastructure constraint on strategic analytics	T4	O4
C40	Rust et al. (2021)	pp.90-96	Metric misalignment creates a self-reinforcing cycle: short-term metrics drive analytics design, which generates clearest signals	Metric misalignment self-reinforcing cycle	Structural metric trap	T4	O4

Code ID	Source	Page/Section	Verbatim Extract / Close Paraphrase	Initial Code Label	Sub-theme	Theme	Obj.
			about short-term performance, which reinforces short-term metric dominance				
C41	Moorman (2020)	pp.35-37	Cultural resistance driven by the creative professional identity of FMCG marketing managers generates pushback against data-driven approaches that technical investment alone cannot overcome	Cultural resistance barrier	Professional identity threat response	T4	O4
C42	Germann, Lilien and Rangaswamy (2013)	pp.120-125	Organisations exhibiting low analytics culture derive significantly lower performance benefits from equivalent analytics investment than those with high analytics culture	Analytics culture performance multiplier	Cultural conditions as performance multiplier	T4	O4
C43	Digital Applied (2026)	Section 3	As AI augmentation shifts the premium from data processing toward strategic integration, organisations concentrating data literacy in specialist roles are compounding the analytical-managerial divide through their own talent decisions	Human capital architecture barrier	Talent structure compounding analytical-managerial divide	T4	O4
C44	Digital Applied (2026)	Section 2	Fully loaded costs per marketer in large FMCG organisations range from USD 180,000 to USD 420,000, with tools and platforms representing 14-18% of that cost in 2026, up from 9% in 2022	Rising analytics cost burden in marketing teams	Financial pressures on analytics human capital investment	T4	O4
C45	IPA Effectiveness Databank (2023)	Chapter 6	Incentive systems rewarding short-term performance actively discourage longer-term brand investment regardless of available analytical evidence, reinforcing metric misalignment structurally	Incentive structure reinforcement of barriers	Governance mechanism perpetuating metric misalignment	T4	O4
C46	Deloitte Insights (2022)	Section 4	The most analytically mature FMCG organisations are distinguished by data integration depth rather than individual tool sophistication , breaking down silos between consumer, retailer, and media data is the primary differentiator	Data integration depth as maturity differentiator	Infrastructure quality vs tool quality distinction	T4	O4

Code ID	Source	Page/Section	Verbatim Extract / Close Paraphrase	Initial Code Label	Sub-theme	Theme	Obj.
C47	Grand View Research (2024)	Market Overview	FMCG is projected to be the highest-spending sector in marketing analytics globally, yet investment scale does not correlate directly with strategic marketing effectiveness gains across the sector	Scale-effectiveness decoupling at sector level	Systemic inadequacy of technical investment alone	T4	O4
C48	Nielsen IQ (2022)	Section 3	The sector underperformed average economic profit by more than 13% in the pre-pandemic decade despite growing analytics investment, reflecting the inadequacy of purely technical investment	Sector economic underperformance despite investment	Aggregate consequence of systemic barriers	T4	O4
C49	Brynjolfsson, Hitt and Kim (2011)	pp.326-334	DDDM adoption showed statistically significant positive associations with firm-level performance, but benefits were contingent on complementary organisational investments in culture and human capital	Contingency of DDDM performance benefits	Organisational conditions as performance prerequisites	T4	O4
C50	Varadarajan (2020)	pp.92-95	Customer information resources advantage requires not just data but the organisational capability to deploy it strategically , without interpretive managerial capability, data generates no competitive advantage	Information resource advantage conditions	Resource-based conditions for analytics competitive advantage	T4	O4

Table A2: Thematic Coding Table , 50 coded entries across four themes. T1=Theme One (Operational Application); T2=Theme Two (Marketing Effectiveness); T3=Theme Three (Managerial Judgement); T4=Theme Four (Structural Barriers). Codes generated through Phase 2 of Braun and Clarke (2006; 2019) six-phase framework.

Appendix 2b: Research Memos

Research memos document the analytical reasoning behind each coding decision, including the significance of the passage, why it was coded as it was, and how it contributes to the theme-level argument. These memos constitute the researcher's reflexive analytical journal referenced in Section 3.5 and provide the evidence that the analytical process was interpretive and systematic rather than impressionistic.

Code ID	Source	Research Memo , Analytical Significance
C01	Wedel and Kannan (2016)	Foundational taxonomy confirming that FMCG analytics applications are by design operational, not strategic. Establishes the baseline from which the operational-strategic gap is measured.
C02	McKinsey and Company (2021)	Challenges the assumption that DDDM deployment is an irreversible capability. Demonstrates that analytics is deployed selectively and conditionally, not systematically.
C03	Nielsen IQ (2022)	Structural evidence that the FMCG analytics capacity is consumed by operational data management, leaving minimal time for strategic insight generation.
C04	Davenport (2023)	Critical theoretical distinction. Analytics tools are architecturally designed for optimisation, not transformation. This makes the operational concentration of DDDM not a choice but a structural inevitability given current tool design.
C05	Davenport and Harris (2017)	Identifies the gap between the theoretical ideal of analytics-driven competition and the operational reality of selective, siloed deployment across FMCG functions.
C06	Consumer Goods Technology (2025)	Concrete case evidence from a major FMCG operator showing that high operational analytics maturity coexists with low strategic analytics application within the same organisation.
C07	McKinsey and Company (2023)	Corroborates the operational concentration finding from an industry intelligence perspective. The value realisation challenge is named here explicitly.
C08	Nielsen IQ (2023)	The 78% figure is frequently cited as evidence of widespread adoption. This entry documents the critical qualification: planning in this context means operational planning, not strategic planning.
C09	Unilever (2024)	Positive case evidence demonstrating that the operational concentration pattern can be overcome through deliberate strategic commitment. Qualifies the theme's central finding.
C10	PepsiCo (2024)	Corroborates Unilever evidence. The key qualifier is the sustained multi-year investment required. This is not achievable through incremental analytics tool improvement alone.
C11	SChain Advisory (2024)	Documents the widespread recognition of DDDM importance alongside the persistent failure to integrate it effectively. Corroborates the operational concentration finding from a practitioner perspective.
C12	Chowdhury and Nath (2024)	Documents a specific mechanism through which operational DDDM concentration is maintained: even when analytics data is available, it is used to validate rather than generate strategic decisions.
C13	Grand View Research (2024)	Critical aggregate evidence that the operational concentration of DDDM is producing a paradox: massive analytics investment without commensurate strategic effectiveness improvement.
C14	Sterling Choice (2024)	Establishes that operational analytics does generate measurable performance advantages. The question is whether these advantages extend to the strategic dimensions examined in Theme Two.

Code ID	Source	Research Memo , Analytical Significance
C15	Germann, Lilien and Rangaswamy (2013)	Identifies the key moderating variable explaining selective effectiveness impact. High integration depth enables broader effectiveness improvements; low integration depth limits impact to short-term metrics.
C16	Binet and Field (2013)	The study's most analytically provocative source. Documents the structural paradox at the heart of DDDM in FMCG: analytics success on measurable dimensions actively damages performance on strategically important ones.
C17	Kantar (2024)	This finding requires careful interpretation: the correlation most plausibly reflects broader organisational sophistication rather than a direct causal effect of analytics on brand equity.
C18	Srinivasan and Hanssens (2022)	Financial evidence of the long-term cost of short-term analytics optimisation. Directly corroborates Binet and Field with shareholder value data.
C19	Trace Brand Building (2024)	The most concrete financial evidence in the corpus. Quantifies the business consequences of analytics-justified short-termism with precise valuation data. The primary case evidence for the measurement trap.
C20	Ambler and Roberts (2008)	Establishes the structural measurement bias that predates and enables the measurement trap. The problem is not that organisations do not know how to measure brand equity , it is that they choose not to.
C21	Davcik and Sharma (2022)	Resource-based view theoretical grounding for the measurement trap finding. The bias is structural and framework-level, not individual or idiosyncratic.
C22	IPA Effectiveness Databank (2023)	IPA Databank evidence demonstrating that short-term metric success actively conceals the progressive erosion of long-term competitive advantage.
C23	Rust et al. (2021)	Connects the measurement trap to the incentive architecture that sustains it. Even when analytical evidence supports brand investment, incentive systems override it.
C24	Analytic Partners (2023)	Demonstrates the signal asymmetry at the heart of the measurement trap: analytics generates stronger signals about the less effective marketing approach (performance marketing).
C25	Moorman (2020)	Most consistently documented pattern in the corpus. Establishes that analytics frequently functions as post-hoc justification rather than decision input , the distinction that reframes the managerial judgement finding.
C26	Moorman and Day (2016)	Foundational theoretical grounding for the constitutive managerial judgement finding. The organisation, not the individual manager, is the primary unit of analytical failure or success.
C27	Lim (2021)	Theoretical grounding for the selective engagement pattern. Bounded rationality explains why data non-use is rational, not irrational, under FMCG commercial pressures.
C28	Sivarajah et al. (2017)	Provides a structured account of the mechanisms through which managerial judgement is distorted in DDDM. Moves the analysis from description to mechanism.
C29	Chowdhury and Nath (2024)	Documents the intuition override pattern empirically in FMCG context. Critically, frames this as rational rather than irrational , the override is a response to inadequate analytical output quality or accessibility.
C30	Davenport and Harris (2017)	Identifies analytical leadership as the key organisational lever for converting distorting managerial judgement into enabling managerial judgement. Links to the comparative company evidence.

Code ID	Source	Research Memo , Analytical Significance
C31	Varadarajan (2020)	Resource-based view theoretical grounding for the constitutive managerial judgement finding. Analytics is a resource; managerial interpretive capacity is what converts it into competitive advantage.
C32	Unilever (2024)	Positive case evidence. Unilever's performance advantage derives from deliberate organisational design that ensures analytical outputs are accessible and actionable for senior judgment.
C33	PepsiCo (2024)	Cultural evidence from PepsiCo leadership. The explicit philosophical framing of analytics as complementary to rather than substituting for judgment is the key cultural differentiator.
C34	McKinsey and Company (2021)	Evidence from post-pandemic period. Leadership quality in integrating analytics with judgment is the differentiating variable between organisations that maintain DDDM commitment and those that retreat.
C35	Trace Brand Building (2024)	The counterfactual case evidence. Documents precisely the post-hoc rationalisation dynamic identified in C25, with measurable competitive failure consequences.
C36	Germann, Lilien and Rangaswamy (2013)	Connects the analytics integration depth finding from Theme Two to the managerial judgement finding. Integration depth is the structural variable that determines whether judgement is enabling or distorting.
C37	Moorman and Day (2016)	Primary structural barrier. Foundational identification of the architectural problem that makes analytics investment insufficient alone for strategic impact.
C38	McKinsey and Company (2023)	Names and characterises the aggregate consequence of the structural barriers. The value realisation challenge is the sector-level expression of the analytical-managerial divide.
C39	Sivarajah et al. (2017)	Third structural barrier. Without data integration, the comprehensive cross-functional views required for strategic DDDM are architecturally impossible regardless of analytical tool sophistication.
C40	Rust et al. (2021)	Second structural barrier. The circular logic of metric misalignment makes it self-perpetuating. Breaking the cycle requires measurement philosophy intervention, not analytics tool improvement.
C41	Moorman (2020)	Fourth barrier. Cultural resistance is the most fundamental because it determines whether organisations can address the other barriers. It is not irrational , it is a rational defence of professional identity.
C42	Germann, Lilien and Rangaswamy (2013)	Quantitative evidence that the cultural barrier has direct performance consequences. Analytics culture is as important as analytics capability in determining business impact.
C43	Digital Applied (2026)	Fifth and most undertheorised barrier. The human capital architecture of analytics teams is itself a structural barrier when analytical literacy is concentrated rather than distributed.
C44	Digital Applied (2026)	Quantifies the financial scale of the human capital challenge. The rising tool cost proportion means organisations must invest more without necessarily developing the strategic integration capability required.
C45	IPA Effectiveness Databank (2023)	Links the metric misalignment barrier to the governance and incentive architecture that sustains it. The problem is not a lack of analytical evidence but a reward system that overrides it.
C46	Deloitte Insights (2022)	Reframes what analytics maturity means: depth of data integration, not sophistication of individual tools. Directly addresses the data fragmentation barrier.

Code ID	Source	Research Memo , Analytical Significance
C47	Grand View Research (2024)	Sector-level evidence that technical investment without systemic transformation is insufficient. The highest analytics spending sector continues to underperform on strategic effectiveness.
C48	Nielsen IQ (2022)	The most important aggregate evidence. A decade of analytics investment produced sector-level economic underperformance. This is the business consequence of the systemic barrier pattern identified in this theme.
C49	Brynjolfsson, Hitt and Kim (2011)	Early evidence establishing that DDDM benefits are contingent rather than automatic. The organisational conditions (culture, human capital) are as important as the technology investment.
C50	Varadarajan (2020)	Resource-based view closure for Theme Four. The systematic identification of barriers explains why analytics investment without organisational transformation fails to generate competitive advantage.

Table A2b: Research Memos, analytical significance of each of the 50 coded entries, constituting the reflexive analytical journal for the thematic analysis process.

Appendix 3: Codebook

The codebook defines all 16 codes used across the 50 coded entries. For each code, the codebook provides a formal definition, the include criteria specifying what counts as evidence for the code, and an example coded entry. The codebook was developed iteratively during Phase 2 and refined during Phase 3 (Searching for Themes) of the Braun and Clarke (2006; 2019) framework.

Code Label	Definition	Include Criteria	Example Entry
Operational taxonomy of analytics	Passages identifying or categorising FMCG analytics applications as primarily operational or tactical in scope	Explicit categorisations of analytics use cases as promotional, media, demand forecasting, supply chain	C01 , Wedel and Kannan taxonomy
Contingency of DDDM deployment	Evidence that analytics deployment is conditional, reversible, or retreated from under pressure	Cases of organisations retreating from data-driven approaches or conditionally adopting them	C02 , pandemic retreat to mass marketing
Low-value analytical time dominance	Evidence that significant analytical capacity is consumed by data structuring, cleaning, or low-value tasks	Percentages or descriptions of analytical time allocation to non-insight work	C03 , 80% low-value task time
Optimisation vs transformation distinction	Conceptual or empirical distinction between analytics used for process optimisation versus strategic transformation	Explicit distinction between efficiency-oriented and transformation-oriented analytics	C04 , Davenport optimisation tool characterisation
Post-hoc rationalisation pattern	Evidence of managers using analytics to legitimise decisions already made rather than as decision inputs	Descriptions of data as legitimisation resource, confirmation bias in data use, reverse engineering of analytical justification	C25 , Moorman legitimisation resource
Analytical-managerial divide	Structural separation of analytics capability from marketing decision authority within organisations	Descriptions of analytics teams structurally separate from decision-makers, coordination failures	C37 , Moorman and Day structural barrier
Measurement trap paradox	The structural paradox whereby optimised analytics generates strong signals about strategically secondary dimensions	Evidence that analytically optimised activities undermine long-term performance; short-term vs long-term trade-offs	C16 , Binet and Field paradox
Bounded rationality in FMCG marketing	Evidence of cognitive or informational constraints shaping how managers engage with analytical outputs	Time pressure, cognitive overload, selective attention,	C27 , Lim bounded rationality application

Code Label	Definition	Include Criteria	Example Entry
		commercial pressure effects on data use	
Metric misalignment self-reinforcing cycle	The circular dynamic whereby short-term metrics drive analytics design which reinforces short-term metric dominance	Descriptions of incentive-metric-analytics circular reinforcement	C40 , Rust et al. self-reinforcing cycle
Analytics integration depth moderator	The extent to which analytics insight is genuinely embedded in decision processes rather than delivered as standalone reports	High/low integration depth comparisons, embedded vs report-based analytics distinctions	C15 , Germann et al. integration depth
Data fragmentation barrier	The structural fragmentation of data across separate organisational systems preventing comprehensive strategic views	Data silo descriptions, integration failures, fragmented data infrastructure	C39 , Sivarajah et al. fragmentation
Cultural resistance barrier	Professional identity-based resistance to data-driven approaches from FMCG marketing managers	Creative identity threats, intuition over data preferences, analytics pushback	C41 , Moorman cultural resistance
Human capital architecture barrier	The structural concentration of analytical literacy in specialist roles rather than distribution across marketing generalists	Specialist-generalist divides, data literacy distribution issues, talent architecture descriptions	C43 , Digital Applied talent structure
Analytical leadership enabling condition	Senior leader behaviour modelling data-informed decisions and creating cultural conditions for analytics integration	Leader analytics modelling, analytical leadership concept, top-down analytics culture creation	C30 , Davenport analytical leadership
High-DDDM strategic extension	Evidence of organisations successfully extending DDDM beyond operational domains into strategic decision-making	Portfolio analytics, brand investment decisions informed by data, strategic acquisition analytics	C09 , Unilever Power Brand strategy
Quantified business consequences of misaligned analytics	Specific financial or business performance evidence of the costs of analytics-justified short-termism	Valuation discounts, market share losses, revenue underperformance linked to analytics misalignment	C19 , Trace Brand Building 40% valuation discount

Table A3: Codebook , 16 codes used in the thematic analysis, with definitions, inclusion criteria, and illustrative examples. Codes are not exhaustive categories but analytical labels applied to passages on the basis of their contribution to the research questions.

Appendix 4: Theme Development Record

The theme development record documents how the 50 initial codes clustered into the four final themes. It shows the contributing codes, the supporting source evidence, and the analytical rationale for each theme. This record corresponds to Phases 3 and 4 of the Braun and Clarke (2006; 2019) framework and constitutes the evidence that thematic development was systematic and analytically grounded rather than arbitrary.

Theme Name	Obj.	Contributing Codes	Supporting Sources	Analytical Rationale
Theme 1: The Operational Application of DDDM in FMCG Marketing	O1	Operational taxonomy of analytics; Contingency of DDDM deployment; Low-value analytical time dominance; Optimisation vs transformation distinction; Gap between analytics aspiration and reality; Operational-strategic asymmetry within organisations; Selective deployment in isolated functions; Perception-reality gap in analytics centrality; High-DDDM strategic extension; Sustained strategic analytics investment; Implementation gap acknowledgement; Post-hoc validation pattern; Investment-effectiveness decoupling; Operational performance advantages documented	C01-C14 drawing on: Wedel and Kannan (2016); Nielsen IQ (2022); Nielsen IQ (2023); Davenport (2023); McKinsey (2021; 2023); Consumer Goods Technology (2025); Unilever (2024); PepsiCo (2024); Sterling Choice (2024); SChain Advisory (2024); Chowdhury and Nath (2024); Grand View Research (2024)	These 14 codes clustered around a consistent pattern: analytics in FMCG is architecturally and organisationally concentrated in operational-tactical functions. The pattern was robust across academic literature, industry intelligence, and company performance data. The tension between the operational concentration documented in C01-C13 and the strategic extension evidence in C09-C10 sharpened rather than undermined the theme: it demonstrated that operational concentration is a choice, not an inevitability.
Theme 2: DDDM and the Dimensions of Marketing Effectiveness	O2	Analytics integration depth moderator; Measurement trap paradox; Analytics-brand equity correlation; Promotional efficiency vs brand investment trade-off; Quantified business consequences of misaligned analytics; Strategic measurement default to convenience; Structural measurement bias; Short-term activation declining returns; Incentive structure reinforcement of short-termism; Brand vs performance marketing evidence	C15-C24 drawing on: Germann et al. (2013); Binet and Field (2013); Kantar (2024); Srinivasan and Hanssens (2022); Trace Brand Building (2024); Ambler and Roberts (2008); Davcik and Sharma (2022); IPA (2023); Rust et al. (2021); Analytic Partners (2023)	These 10 codes converged on a structural paradox: the effectiveness dimensions most amenable to DDDM optimisation are precisely those whose overemphasis most damages long-term competitive advantage. The mid-sized manufacturer case (C19) provided concrete financial quantification of this paradox. The tension between C17 (analytics-brand equity correlation) and the weight of evidence required the researcher to offer a

Theme Name	Obj.	Contributing Codes	Supporting Sources	Analytical Rationale
				qualified interpretation, supporting methodological rigour.
Theme 3: Managerial Judgement as Mediator	O3	Post-hoc rationalisation pattern; Organisational context as primary moderator; Bounded rationality in FMCG marketing; Three-category judgement engagement challenge; Category intuition override of model recommendations; Analytical leadership enabling condition; Managerial judgement as constitutive interpretive capacity; Leadership-analytics integration design; Complementarity philosophy of data and judgment; Analytical leadership under volatility; Post-hoc rationalisation leading to competitive failure; Integration depth determines judgement-analytics relationship	C25-C36 drawing on: Moorman (2020); Moorman and Day (2016); Lim (2021); Sivarajah et al. (2017); Chowdhury and Nath (2024); Davenport and Harris (2017); Varadarajan (2020); Unilever (2024); PepsiCo (2024); McKinsey (2021); Trace Brand Building (2024); Germann et al. (2013)	These 12 codes converged across methodologically diverse source types , peer-reviewed, industry, and company performance data , on the finding that managerial judgement is constitutive rather than peripheral. The convergence across source types significantly strengthens the analytical claim. The critical tension acknowledged in the findings (judgement as both distorting and enabling) is grounded in coded evidence from C26 and C30-C34 respectively.
Theme 4: Structural and Cultural Barriers to Strategic DDDM	O4	Analytical-managerial divide; Value realisation challenge mechanism; Data fragmentation barrier; Metric misalignment self-reinforcing cycle; Cultural resistance barrier; Analytics culture performance multiplier; Human capital architecture barrier; Rising analytics cost burden; Incentive structure reinforcement of barriers; Data integration depth as maturity differentiator; Scale-effectiveness decoupling at sector level; Sector economic underperformance despite investment; Contingency of DDDM performance benefits; Information resource advantage conditions	C37-C50 drawing on: Moorman and Day (2016); McKinsey (2023); Sivarajah et al. (2017); Rust et al. (2021); Moorman (2020); Germann et al. (2013); Digital Applied (2026); IPA (2023); Deloitte (2022); Grand View Research (2024); Nielsen IQ (2022); Brynjolfsson et al. (2011); Varadarajan (2020)	These 14 codes cluster around five mutually reinforcing barrier categories. The systemic interpretation , that barriers constitute a mutually reinforcing pattern rather than a list of independent problems , is the researcher's analytical construction, grounded in the pattern of co-occurrence across the corpus. The key analytical move is from listing barriers to identifying their systemic interdependence. The weight of evidence from C48 (13% sector underperformance) provides the business consequence quantification that validates the systemic interpretation.

Table A4: Theme Development Record , showing the progression from coded entries to sub-themes to final themes across the four research objectives. Produced during Phases 3-5 of the Braun and Clarke (2006; 2019) six-phase reflexive thematic analysis framework.

Appendix 5: Ethical Approval

Ethical approval was obtained from Newcastle University Business School prior to data collection commencement, in accordance with the University's Policy and PGT Dissertation Ethics Approval Procedure. This study used exclusively publicly available secondary data and involved no primary data collection with human participants.

5.1 Business School - Ethical approval form for student projects

SECTION 1: Applicant Details

Student Name and Number:	Shazia Shehzeen Prova
Student Email Address:	S.S.Prova2@newcastle.ac.uk
Programme of Study:	MSc Business Analytics
Module Code and Name:	NBS8062: Research Methods
Supervisor Name:	Dr. Jacqueline Adamson
Supervisor Email Address:	Jackie.Adamson@newcastle.ac.uk

SECTION 2: Project Details

Project Title:	How Data-Driven Decision-Making Influences Marketing Effectiveness in the FMCG Industry: A Critical Qualitative Study
Project Synopsis: *Please describe here what your planned research is about, why it is important and how you seek to conduct it.*	Data-driven decision making (DDDM) has gained significant prominence in the Fast-Moving Consumer Goods (FMCG) industry, with organisations investing heavily in analytics platforms to improve marketing outcomes. Despite this growing adoption, the relationship between DDDM and marketing effectiveness remains insufficiently examined from a managerial perspective. This study adopts an interpretivist, qualitative approach to critically investigate that relationship. Secondary data drawn from peer-reviewed journals, industry reports, and published FMCG case studies will be collected and analysed using reflexive thematic analysis via NVivo 14.

SECTION 3: Project Type

	YES	**NO**
This research involves primary research.	☐	☒
This research involves secondary research.	☒	☐

*If you ticked **YES**, please complete the remainder of the form and append a copy of the information sheet and consent form for your research.*

*If you ticked **NO**, please sign the form on the last page and submit it to your supervisor for sign-off.*

	YES	NO
This research involves travel.	☐	☒

*If you ticked **YES**, please complete and append the risk assessment form 'travel'.*

*If you ticked **NO**, please complete and append the risk assessment form 'no travel'.*

SECTION 4: NHS, Health & Social Care: Facilities, Staff & Patients

	YES	NO
"Will the study involve participants recruited by virtue of being service users, their dependents, their carers' or human tissues or the use of NHS & Health / Social Care Facilities?	☐	☒

*If you ticked **YES**, Please discuss alternative research questions / settings with your supervisor as a matter of urgency. The School does not support student projects involving patients of the NHS (or another health system in the UK or abroad) or their relatives.*

*If you ticked **NO**, please go to Section 5.*

SECTION 5: Human Participants in a Non-Clinical Setting

	YES	NO
Does the research involve human participants, e.g. use of questionnaires, interviews, focus groups, observation, surveys or lab-based studies involving human participants? (If you are unsure please tick 'Yes' and complete the sub-questions.)	☐	☒

*If you ticked **YES**, please complete the sub-questions below and provide details.*

*If you ticked **NO**, please go to Section 6.*

Does the study involve any of the following?	YES	NO
a. The study involves children, or other vulnerable groups (inc. DBS should students want to work in schools), defined in Section 59 of the Safeguarding Vulnerable Adults Act 2006 as those who are relatively or absolutely incapable of protecting their own interests, or those in unequal relationships, e.g. participants who are subordinate to the researcher in a context outside the research[1]?	☐	☒
b. The study requires the co-operation of a gatekeeper for initial access to the groups or individuals to be recruited (e.g. organisational representative, chair / committee of a	☐	☒

[1] Research involving adults (16+) who cannot consent, must have NHS review even if it is not health related.

professional association)? **Please note: gatekeepers are normally required when a case-study approach is taken.**		
c. It is necessary for participants to take part in the study without their knowledge and consent e.g. covert observation of people in non-public places?	☐	☒
d. Deliberately misleading participants in any way?	☐	☒
e. Psychological stress, anxiety, harm or negative consequences beyond that encountered in normal life?	☐	☒
f. Prolonged or repetitive testing i.e., more than 4 hours' commitment or attendance on more than two occasions?	☐	☒
g. Financial inducements (other than reasonable expenses and compensation for time)?	☐	☒

*If you have answered **YES** to any of the sub-questions in Section 5, please provide details about the research, the issues that you have identified and the actions through which you seek to mitigate against them. Particular attention should be given to the procedures by which informed consent (the form should include the student's and supervisor's details to protect both researcher and interviewees) is sought and by which participants can withdraw from the research. **Please append a copy of the information sheet and consent form for your project**. The text box can be increased in size.*

Please note that only low-risk projects are supported. Your supervisor may demand alterations to your research to ensure that it is low risk before being able to sign off this form.

Please continue with the Section 6.

SECTION 6: Data

	YES	NO
Does the research involve the viewing, usage or transfer of Sensitive personal data as defined as by the Terrorism Act or data governed by statute such as the Official Secrets Act, commercial contract or by convention e.g. client confidentiality? (If you are unsure please tick 'Yes' and complete the sub-questions.)	☐	☒

*If you ticked **YES**, please complete the sub-questions below.*

*If you ticked **NO**, please go to Section 7.*

	YES	NO
a. Will the study involve the sharing of sensitive data outside the European Economic Area?	☐	☐

b. Will the study involve the collection or analysis of sensitive data which will be identifiable within the project outputs and could potentially cause harm?	☐	☐
c. Will the study involve the collection or analysis of personal data without explicit consent?	☐	☐
d. Will the study involve the collection, viewing or dissemination of materials which could be considered 'extremist' and or 'terrorism related'?	☐	☐

*If you have answered **YES** to any of the sub-questions in Section 6, there is a presumption that dissertation/project proposals will need individual ethical clearance. This would normally be incorporated in the existing project/dissertation proposal approval process. In addition, ensure that data management protocols are in place to conform with GDPR (2018) legislation. To comply with GDPR – ensure that all data is securely transferred via an encrypted USB when off-site, confidential, anonymous, the right to be forgotten and data will be destroyed post study. Similarly, if students are conducting lone interviews they should conform to the 'Lone Worker' protocol. T*he text box can be increased in size.

Please note that only low-risk projects are supported. Your supervisor may demand alterations to your research to ensure that it is low risk before being able to sign off this form.

Please continue with Section 7.

SECTION 7: International Projects

	YES	NO
Will the research be conducted outside of the European Economic Area (EEA) or will it involve international collaborators outside the EEA?	☐	☒

*If you ticked **YES**, please complete the sub-questions below.*

*If you ticked **NO**, please go to Section 8.*

	YES	NO
a. Will there be an increased risk to researchers and participants where they are working remotely?	☐	☐
b. Will the research involve political sensitivities (e.g. offend specific public interests or the rights and reputations of others)?	☐	☐
c. Will the work undertaken overseas be governed by standards which are NOT equivalent to those in the UK?	☐	☐

*If you have answered **YES** to any of the sub-questions in Section 7, please provide details about the research, the issues that you have identified and the actions through which you seek to mitigate against them. You should abide by the University's 'Lone Worker' protocols in informing family or friends of your whereabouts when collecting data. Particular attention should be given to differing expectations between the UK and the country in question and any risks associated. The text box can be increased in size.*

Please note that only low-risk projects are supported. Your supervisor may demand alterations to your research to ensure that it is low risk before being able to sign off this form.

Please continue with the Section 8.

SECTION 8: Declaration

By signing this declaration I, Shazia Shehzeen Prova: * certify that the information contained in this application is accurate. * certify that the research will be undertaken in line with all appropriate local standards and regulations. * certify that I have discussed this form with my project supervisor and that s/he agrees with the answers given. * will submit a revised form should the project design change after this form has been signed off.	
Name of student:	Shazia Shehzeen Prova
Signature of student:	Shazia Shehzeen Prova
Date:	08-05-2026

SECTION 9: Supervisor sign-off and comments

If any of the sub-questions in Sections 5 to 7 has been answered with "yes", please provide a written justification below as to why the project as a whole is still low-risk. Special attention should be given to the conduct of the research, particularly when human participants are involved, when a gatekeeper is required and/or the research takes place internationally. Details about discussions with the student about key ethical issues and their mitigation should also be provided here.

Name of supervisor:	
Signature of supervisor:	
Date:	

Please attach any risk assessment forms and other applicable documents (such as information sheet and consent form) and submit them for sign-off as advertised in the module guide.

5.2 Risk Assessment – No Travel

Newcastle University Business School Risk Assessment – No Travel etc.		
Module Code and Title	NBS8062: Research Methods	
Project Title	How Data-Driven Decision-Making Influences Marketing Effectiveness in the FMCG Industry: A Critical Qualitative Study	
Project synopsis (can be copied from Ethics form)	Data-driven decision making (DDDM) has gained significant prominence in the Fast-Moving Consumer Goods (FMCG) industry, with organisations investing heavily in analytics platforms to improve marketing outcomes. Despite this growing adoption, the relationship between DDDM and marketing effectiveness remains insufficiently examined from a managerial perspective. This study adopts an interpretivist, qualitative approach to critically investigate that relationship. Secondary data drawn from peer-reviewed journals, industry reports, and published FMCG case studies will be collected and analysed using reflexive thematic analysis via NVivo 14.	
Confirmation	I confirm that no travel outside of the University will be undertaken for the purposes of this research project.	
Student		
Name: Shazia Shehzeen Prova	**Signature:** Shazia Shehzeen Prova	**Date 08-05-2026**
Supervisor		
Name:	**Signature:**	**Date**

5.3 GDPR Data Management Assessment

Student Name: Shazia Shehzeen Prova		
Supervisor: Dr. Jacqueline Adamson		
Title of project: How Data-Driven Decision-Making Influences Marketing Effectiveness in the FMCG Industry: A Critical Qualitative Study		
Documenting:		
	Yes	**No**
I understand that I must retain all my data until after I have graduated	X	
I will provide a methodology that explains how my data was collected	X	
Please explain how you will label and organise your data, records and files		
I will label the data file as 'Dissertation Literature Sources' and retain it on the university's SharePoint. I will not delete the data until after I have graduated.		
Storing:		
	Yes	**No**
I understand that I must keep all data and documentation related to my dissertation on the University server	X	
I will protect personal or sensitive data	X	
Please explain how you will protect personal or sensitive data (if you are not collecting this sort of data, please state this here)		
I will not collect this sort of data. The information sheet and consent form explain this in detail.		
Please explain how you will transfer and store data that is collected with mobile devices (e.g. audio recorder, mobile phone, USB stick, laptop)		
I will not collect this sort of data. The information sheet and consent form explain this in detail.		

Confidentiality, ethics and consent:			
	Yes	**No**	**N/A**
I have completed the ethics assessment	X		
I will follow this assessment throughout my dissertation and seek additional advice from my supervisor as needed	X		
I have gained consent from my participants, including their permission to share data with supervisor/second marker			X
I have anonymised my data to remove identifying/personal information			X
Copyright			
	Yes	**No**	**N/A**
Have you established who owns the copyright in your data?	**X**		
If you are re-using someone else's data sources, have you considered copyright?	**X**		
Please explain how you have accessed any secondary data being used and any associated copyright implications			
I will access the secondary literature via the access provided by the university's library. The university has subscribed to these databases and no copyright issues should be raised.			